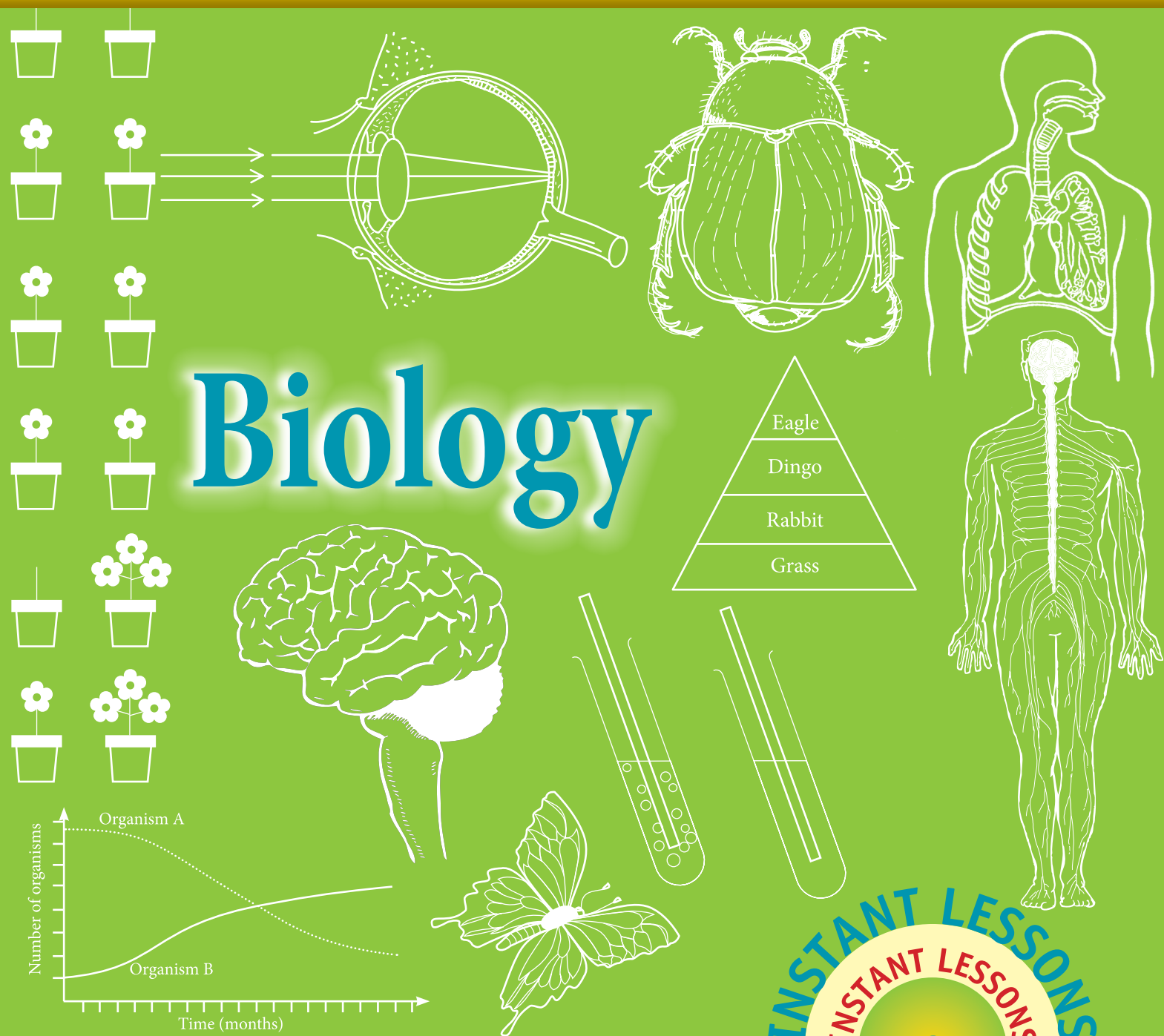


MULTIPLE CHOICE TEST TOPICS



100s of ready-to-use topic questions!

Biology



Peter Gribben



**M U L T I P L E
C H O I C E
T E S T T O P I C S**

1005 of ready-to-use topic questions!

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This book was commissioned by Jennifer Gough, production was by Piper Print Media, and David Dickson supplied the illustrations.

Peter Gribben is currently Head Teacher of Science and Agriculture at Crookwell High School in the Southern Tablelands of New South Wales. He has taught Science at all levels for around 30 years in England, the Middle East and Australia. Peter has written many science books; both conventional textbooks and photocopiable worksheets. He also has considerable experience in preparing examination material. Peter has a special interest in preparing science materials that are accessible to students of all levels.

Peter's personal interests include raising German Shepherds, restoring Minis and reading science fiction.

The publisher would like to thank Peter Soucek for his assistance with the preparation of this series.

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INTRODUCTION

This series of photocopiable test topic books has been written to assist teachers to set tests and exams for students in years 9–10. The pages and questions can also be used:

- ✱ to prepare students for this specific type of assessment in more senior year levels
- ✱ as a diagnostic tool to check students' understanding of a just completed topic
- ✱ for homework, extension and revision.

The questions in this series ask students to recall or remember a fact, make calculations, interpret data from a table, make sense of a diagram, read off values from a graph, forecast what will happen from a graph and extract information from a written passage.

Each question could be answered in the normal test manner or students could:

- ✱ be asked to justify their response
- ✱ describe why the wrong answers are incorrect
- ✱ provide examples they know that illustrate and support the correct answer.

Teachers should advise students how to record the answers – such as circling the best answer or answering on a separate answer grid. A list of helpful tips for answering multiple choice questions has been provided and was written by teachers well used to setting and marking tests and preparing their students for examinations.

An editable version of all the questions is also available on the CD-ROM that is attached to the inside back cover of this title. All material can be edited only in accordance with the user agreement.

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Before the test or exam

- ✱ Make sure you have planned out how much time you have to answer the multiple choice section of the exam and know exactly how many marks this section is worth.
- ✱ Do as many practice multiple choice questions from past exams as you can.
- ✱ Ensure that you know the meaning of common words found in exams such as *all, many, some, sometimes, always, never, none, not, few*. Also know the definitions of key words such as *describe, analyse* and *evaluate*.

Reading test questions

- ✱ Read through the paper, and select the easiest questions to do first. Do the hardest questions last. This gives you confidence to do your best.
- ✱ Read each question and try to answer it without looking at the choices. Then read through the choices. Doing this allows you to make a clearer and more accurate choice.
- ✱ Be alert for grammatical inconsistencies between the question and the potential answers. A choice is nearly always wrong if the question and the answer don't combine to make a grammatically correct sentence.
- ✱ Don't read anything into or out of the question, or assume it has been written to trick you.
- ✱ Some questions are tricky though. For example, *Which of the following is not ...; All of the following except ...* A response may be true, but not the answer to the question set.
- ✱ The wrong answers are called 'distractors' – there's a reason for this name!

Strategies

- ✱ Eliminate answers you know aren't right and be logical in your approach.
- ✱ With the harder questions, delete the answers that are least likely to be right. Your chances of selecting a correct answer increases with every alternative you eliminate.

- ✱ If there is no penalty for guessing, and you cannot work out the right answer, make learned guesses.
- ✱ Never leave a multiple choice question unanswered.
- ✱ Don't be put off a response just because it seems too obvious or too simple.
- ✱ If two responses appear to be equally correct, eliminate the response that appears to be least related to the question being asked.
- ✱ Use a common sense approach to numerical questions. First think out whether the answer will be in 10s, 100s, 1000s or millions.
- ✱ Unless otherwise instructed, you may write your workings out or reasoning on the test paper and this will not be looked at by the marker.

At the end of the test or exam

- ✱ Leave yourself time at the end of the exam to check your answer sheet to see if you have recorded your answers correctly.
- ✱ Do not change answers unless you know that your recorded answers are incorrect.
- ✱ Do not change your answers just because there are too many or too few of a particular letter on your answer sheet.

After you have been given your marked answer sheet

- ✱ Look carefully at the questions you answered incorrectly. If you answered them wrongly because you did not understand the content, then it is 'back to the books'.
- ✱ If you made an error because your multiple choice test technique was poor, then re-read the tips for answering this kind of question and do more practice.
- ✱ If anxiety meant that you did not perform as well as you wanted, remember that sound planning and a good test technique will help you to feel more confident in any exam. Practice boosts confidence and lessens anxiety.

1 The general term for a living thing is:

- ☐ A an organism.
- ☐ B a producer.
- ☐ C a consumer.
- ☐ D an individual.

2 Which of the following living things is the smallest?

- ☐ A Bacterium
- ☐ B Earthworm
- ☐ C Crab
- ☐ D Flea

3 Which of the following is *not* a feature of all living things?

- ☐ A They reproduce.
- ☐ B They evolve.
- ☐ C They are composed of cells.
- ☐ D They photosynthesise.

4 All living things respond to stimuli. Which of the following is an example of a stimulus?

- ☐ A Heat
- ☐ B Pressure
- ☐ C Sunlight
- ☐ D All of the above

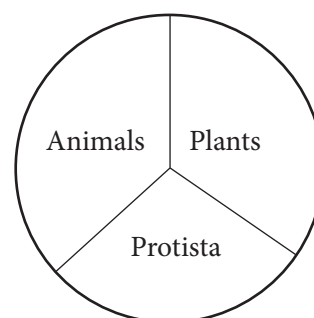
5 One characteristic of living things is that they display *irritability*. This term means that they:

- ☐ A are bad-tempered.
- ☐ B change over time.
- ☐ C respond to stimuli from the environment.
- ☐ D process energy.

6 Scientists classify organisms because:

- ☐ A it makes it easier to identify organisms.
- ☐ B there are only a few organisms that differ from each other.
- ☐ C most discoveries in biology have already been made.
- ☐ D all organisms have characteristics in common.

7 What does the diagram below show?



- ☐ A One way of classifying organisms
- ☐ B The relative numbers of organisms existing in the world
- ☐ C All the possible types of organisms
- ☐ D The different types of animals currently living

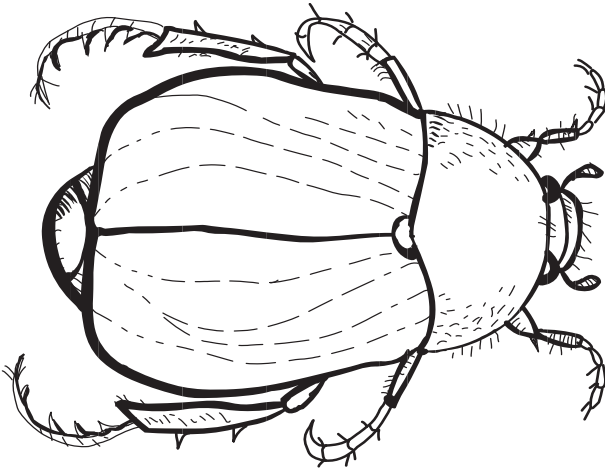
8 Look at the table below.

Set 1	Set 2
apple	stone
worm	iron nail
moss	watch
grass	bicycle
tree	ink
coral	radio

Which is the correct name for Set 1?

- ☐ A Moving
- ☐ B Non-living
- ☐ C Living
- ☐ D Stationary

- 9** The diagram of the insect is shown at 100 times its actual size.



The actual length of this creature is nearest to:

- ☐ A 0.8 mm.
 - ☐ B 8 mm.
 - ☐ C 80 mm.
 - ☐ D 800 mm.
- 10** A vertebrate is an animal which has:
- ☐ A a backbone.
 - ☐ B a hard shell.
 - ☐ C wings.
 - ☐ D fins.
- 11** Echidnas and platypus are classified as:
- ☐ A marsupials.
 - ☐ B eutherians.
 - ☐ C monotremes.
 - ☐ D reptiles.
- 12** To which group does a rose belong?
- ☐ A Vertebrate
 - ☐ B Invertebrate
 - ☐ C Plant
 - ☐ D None of the above

- 13** To which group does a jellyfish belong?

- ☐ A Vertebrate
- ☐ B Invertebrate
- ☐ C Plant
- ☐ D None of the above

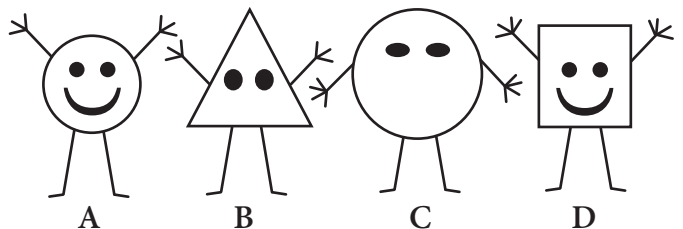
- 14** To which group does a dog belong?

- ☐ A Vertebrate
- ☐ B Invertebrate
- ☐ C Plant
- ☐ D None of the above

- 15** When scientists find a living creature they have not seen before, they use a key to help them classify the organism. Which of the following most closely resembles this type of 'key'?

- ☐ A Metal object used to lock and unlock doors
- ☐ B Book of pressed flowers
- ☐ C A map showing where plants and animals are found in the world
- ☐ D A guide to identifying plants and animals

- 16** The diagram shows four characters.



Use the instructions below to identify Harry.

- | | |
|-------------------|----------|
| 1 Round body | Go to 4. |
| Not round body | Go to 2. |
| 2 Three fingers | Go to 3. |
| Four fingers | Go to 4. |
| 3 Triangular body | Gertie |
| Square body | Iris |
| 4 Mouth | Fred |
| No mouth | Harry |

- ☐ A A ☐ B B ☐ C C ☐ D D

17 Sea urchins and starfish belong to a family of invertebrate animals with spiny skins. Echidnas and porcupines also have spiny skins, but are not included in this same family because they:

- ☐ A cannot swim underwater.
- ☐ B sometimes eat fish.
- ☐ C look after their young.
- ☐ D have backbones.

18 Lobsters, crabs, insects, spiders and centipedes all belong to a group of animals called 'arthropods'. What do these animals all have in common?

- ☐ A They have six legs.
- ☐ B They have jointed legs.
- ☐ C They live in the sea.
- ☐ D They are nice to eat.

19 Fish, amphibians and reptiles have in common the fact that they:

- ☐ A are homeothermic ('warm-blooded').
- ☐ B live in water.
- ☐ C are vertebrates.
- ☐ D are covered in scales.

20 What is the main difference between plants and animals?

- ☐ A Plants make their own food; animals need a source of food.
- ☐ B Animals reproduce, plants don't.
- ☐ C Animals need water, plants don't need water.
- ☐ D Plants are more complicated than animals.

21 Organisms can be grouped into five major kingdoms. These are:

- ☐ A mammals, fish, birds, amphibians and reptiles.
- ☐ B animals, fungi, monera, plants and protists.
- ☐ C animals, fungi, insects, plants and protists.
- ☐ D animals, plants, insects, birds and reptiles.

22 Kingdoms themselves can be sub-divided. The sequence of classification following kingdoms is:

- ☐ A species, phylum, class, order, family, genus.
- ☐ B class, order, family, genus, species, phylum.
- ☐ C order, family, genus, phylum, class, species.
- ☐ D phylum, class, order, family, genus, species.

23 Which row in the table correctly shows how these organisms obtain their nutrition?

	Animals	Plants	Fungi
A	ingest food	photosynthesise	photosynthesise
B	absorb food	ingest food	photosynthesis
C	ingest food	absorb food	absorb food
D	ingest food	photosynthesise	absorb food

- ☐ A A ☐ B B ☐ C C ☐ D D

24 Which column in the table best describes the appearance of the skin on each type of animal?

	A	B	C	D
Mammal	covered with hair or fur	dry and no hair	covered with hair or fur	dry and scales
Bird	covered with hair or fur	covered with feathers	covered with feathers	covered with feathers
Reptile	dry and scales	wet and scales	scales	smooth
Amphibian	wet and no hair	wet and scales	wet and no hair	dry and scales
Fish	wet	smooth	scales	scales

- ☐ A ☐ B ☐ C ☐ D

25 Which column in the table best describes the means of reproduction in each type of animal?

	A	B	C	D
Mammal	females lay eggs with hard shells	females bear live young	females bear live young	females bear live young
Bird	females bear live young	females lay eggs with no shells	females lay eggs with hard shells	females lay eggs with soft shells
Reptile	females lay eggs with no shells	females lay eggs with soft shells	females lay eggs with soft shells	females lay eggs with hard shells
Amphibian	females lay eggs with no shells	eggs are fertilised in water	eggs are fertilised in water	eggs are fertilised in water
Fish	females bear live young	eggs are fertilised in water	eggs are fertilised in water	females lay eggs with no shells

- ☐ A ☐ B ☐ C ☐ D

26 *Plants or animals having common attributes and a common name that can breed and produce viable offspring.*

This defines:

- ☐ A organisms.
☐ B individuals.
☐ C a species.
☐ D a population.

27 *Biological classification is a method by which biologists group and categorise organisms.*

This describes:

- ☐ A entomology.
☐ B taxonomy.
☐ C zoology.
☐ D physiology.

28 Botany is the study of:

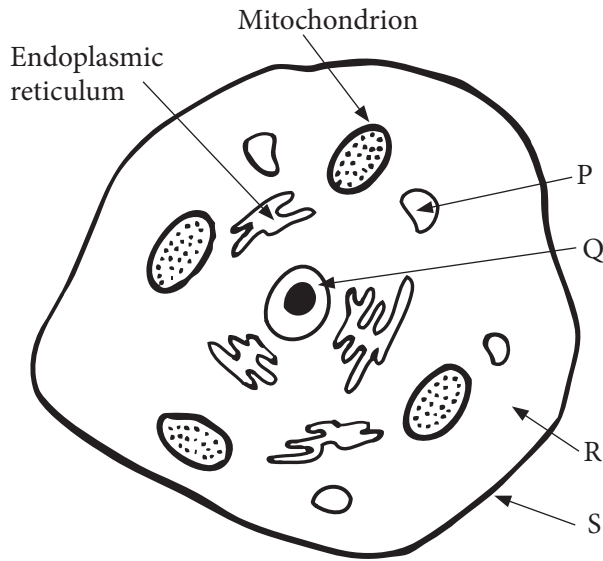
- ☐ A animals.
☐ B plants.
☐ C insects.
☐ D fossils.

29 *Lived from 1707 to 1778, his great work was the 'Systema Naturae'. He used five ranks to classify organisms: class, order, genus, species and variety.*

Who does this describe?

- ☐ A Jean-Baptiste Lamarck
☐ B Gregor Mendel
☐ C Alfred Russell Wallace
☐ D Carl Linnaeus

Questions 1–4 refer to the following diagram of an animal cell.



1 The part labelled 'P' is the:

- ☐ A cytoplasm.
- ☐ B membrane.
- ☐ C nucleus.
- ☐ D vacuole.

2 The part labelled 'Q' is the:

- ☐ A cytoplasm.
- ☐ B membrane.
- ☐ C nucleus.
- ☐ D vacuole.

3 The part labelled 'R' is the:

- ☐ A cytoplasm.
- ☐ B membrane.
- ☐ C nucleus.
- ☐ D vacuole.

4 The part labelled 'S' is the:

- ☐ A cytoplasm.
- ☐ B membrane.
- ☐ C nucleus.
- ☐ D vacuole.

5 Which row of the table matches the correct description with each part of the cell?

	Cytoplasm	Membrane	Nucleus	Vacuole
A	jelly-like	surrounds the cell	controls the cell	stores food
B	provides energy	controls the cell	manufactures dna	transport system
C	mainly liquid	keeps cell rigid	stores food	synthesise protein
D	stores food	surrounds the cell	controls the cell	site of respiration

- ☐ A A ☐ B B ☐ C C ☐ D D

6 The parts that make up the cell are called:

- ☐ A organs.
- ☐ B organelles.
- ☐ C components.
- ☐ D molecules.

7 There are some animals which are:

- ☐ A not made of cells.
- ☐ B made of only one cell.
- ☐ C smaller than cells.
- ☐ D too big to have any cells.

8 A group of special cells in the body that are of the same shape and size and do the same job is called:

- ☐ A a tissue.
- ☐ B an animal.
- ☐ C an embryo.
- ☐ D a nucleus.

9 Body cells *not* involved in reproduction are called:

- ☐ A somatic cells.
- ☐ B germ cells.
- ☐ C sex cells.
- ☐ D zygotes.

10 Cell division is used in:

- ☐ A growth, repair and reproduction.
- ☐ B reproduction and homeostasis.
- ☐ C growth only.
- ☐ D reproduction only.

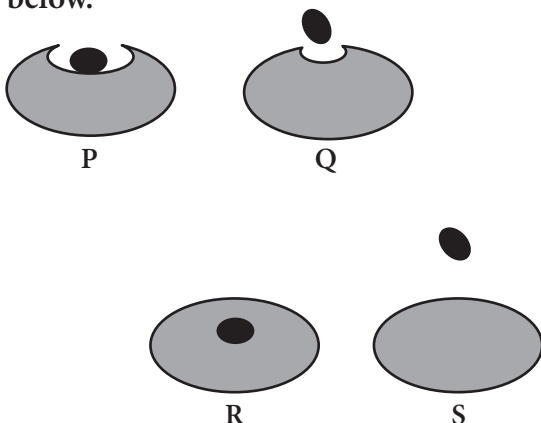
11 Mitosis:

- ☐ A is a form of binary fission.
- ☐ B produces new cells for growing organisms.
- ☐ C replaces cells which have been damaged or have died.
- ☐ D is all of the above.

12 Meiosis:

- ☐ A produces daughter cells with the same number of chromosomes as parent cells.
- ☐ B produces sex cells.
- ☐ C replaces cells which have been damaged or have died.
- ☐ D only occurs in animal cells.

13 Phagocytes are special cells that engulf solid particles such as bacteria. This process is shown in the four diagrams below.



What is the correct sequence for this process?

- ☐ A P, Q, R and then S
- ☐ B Q, P, R and then S
- ☐ C S, Q, P and then R
- ☐ D R, P, Q and then S

14 The part of a microscope that we pick up when we move it is the:

- ☐ A arm.
- ☐ B mirror.
- ☐ C stage.
- ☐ D adjustment.

15 When using a microscope, to put an object into focus we use the:

- ☐ A arm.
- ☐ B mirror.
- ☐ C stage.
- ☐ D adjustment.

16 When using a microscope, to put more light on an object we use the:

- ☐ A arm.
- ☐ B mirror.
- ☐ C stage.
- ☐ D adjustment.

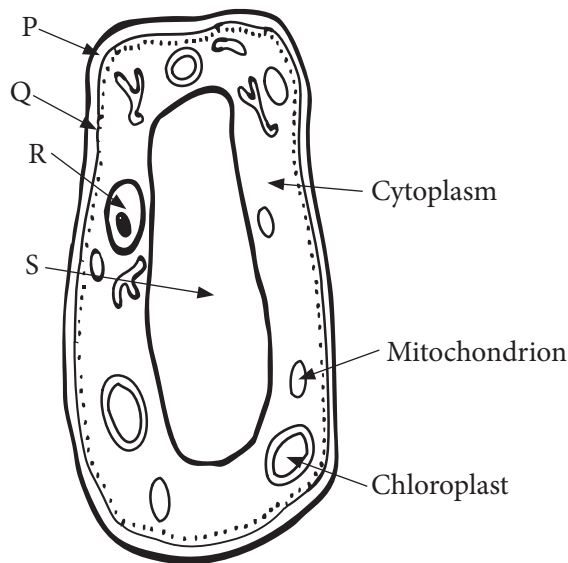
17 When using a microscope, we place the object to be viewed on the:

- ☐ A arm.
- ☐ B mirror.
- ☐ C stage.
- ☐ D adjustment.

18 Which of the following statements about animal cells is true?

- ☐ A They contain chloroplasts.
- ☐ B The outer part is made of cellulose.
- ☐ C They are always multicellular.
- ☐ D The cell's contents are called the protoplasm.

Questions 1–4 refer to the simplified diagram below of a plant cell.



1 The part labelled 'P' is the:

- ☐ A cell membrane.
- ☐ B cell wall.
- ☐ C nucleus.
- ☐ D vacuole.

2 The part labelled 'Q' is the:

- ☐ A cell membrane.
- ☐ B cell wall.
- ☐ C nucleus.
- ☐ D vacuole.

3 The part labelled 'R' is the:

- ☐ A cell membrane.
- ☐ B cell wall.
- ☐ C nucleus.
- ☐ D vacuole.

4 The part labelled 'S' is the:

- ☐ A cell membrane.
- ☐ B cell wall.
- ☐ C nucleus.
- ☐ D vacuole.

5 In which row of the table is the correct description matched with each part of the cell?

	Cell membrane	Cell wall	Nucleus	Vacuole
A	jelly-like	surrounds the cell	controls the cell	stores food
B	skin-like	controls the cell	manufactures dna	rigid structure
C	semi-permeable	keeps the cell rigid	controls the cell	filled with liquid
D	keeps cell rigid	surrounds the cell	controls the cell	contains chlorophyll

- ☐ A A ☐ B B ☐ C C ☐ D D

6 Which one of the following is found in a plant cell but not in an animal cell?

- ☐ A Cell membrane
- ☐ B Mitochondrion
- ☐ C Chloroplast
- ☐ D Nucleus

7 Where does photosynthesis take place in the cell?

- ☐ A All through the cell
- ☐ B In the mitochondria
- ☐ C In the chloroplasts
- ☐ D In the nucleus

8 The cell wall of plants is mainly made of:

- ☐ A cellulose.
- ☐ B protein.
- ☐ C water.
- ☐ D fat.

1 Plants are called 'producers' because they:

- ☐ A make their own food.
- ☐ B produce more plants.
- ☐ C are eaten by other animals.
- ☐ D grow all over the world.

2 A green-coloured piece of material has been found on a beach. To decide if it is a plant, the most scientific thing to do would be to:

- ☐ A prod it to see if it responds to touch.
- ☐ B state that it must be a plant because it is green.
- ☐ C use a microscope to look at any cells present.
- ☐ D cut off a piece, place it in soil and see if it grows.

3 In what form do most plants store food?

- ☐ A Bark
- ☐ B Starch
- ☐ C Sugar
- ☐ D Glycogen.

4 For most seeds to germinate, they need:

- ☐ A oxygen, water and warmth.
- ☐ B carbon dioxide, water and warmth.
- ☐ C oxygen, sunlight and dryness.
- ☐ D carbon dioxide, water and sunlight.

5 'Flora' is the scientific name for:

- ☐ A animals.
- ☐ B insects.
- ☐ C plants.
- ☐ D medicines.

6 Which row of the table shows the correct equation for photosynthesis?

A	carbon + water → glucose + oxygen
B	carbon dioxide + nitrogen → glucose + oxygen
C	nitrogen + water → glucose + oxygen
D	carbon dioxide + water → glucose + oxygen

- ☐ A A
- ☐ B B
- ☐ C C
- ☐ D D

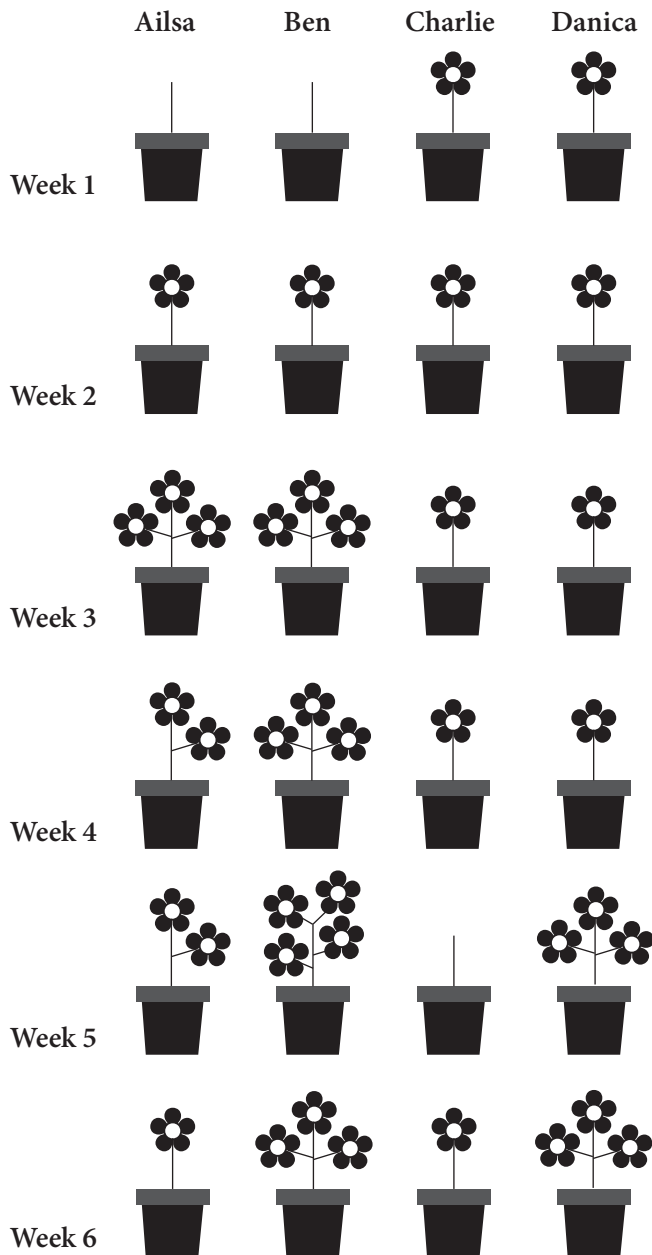
7 Which of the following shows the correct function for each part of the plant?

- ☐ A Stem: transports fluids
Leaf: brings water into plant
Flower: site of photosynthesis
Roots: support plant
- ☐ B Stem: stores nutrients
Leaf: site of gas exchange
Flower: organ of reproduction
Roots: lose water to soil
- ☐ C Stem: supports plant
Leaf: site of photosynthesis
Flower: organ of reproduction
Roots: bring water into plant
- ☐ D Stem: produces new living tissue
Leaf: attracts insects
Flower: organ of reproduction
Roots: anchor plant to ground

8 Which of the following statements about plants is true?

- ☐ A They are always grown from seeds.
- ☐ B They can reproduce sexually or asexually.
- ☐ C They are always multicellular.
- ☐ D The study of plants is called zoology.

- 9** Four students set up flowering plants in their classroom. Each week they counted the number of flowers on their plants. The results are shown in the diagram.



Whose plant had the greatest change in number of flowers between weeks 4 and 5?

- ☐ A Ailsa
- ☐ B Ben
- ☐ C Charlie
- ☐ D Danica

- 10** Stomates are:

- ☐ A where food is digested in plants.
- ☐ B part of a plant's roots.
- ☐ C holes in the leaves.
- ☐ D part of a plant's stem.

- 11** The substance in plants that is required for photosynthesis and makes plants green in colour is:

- ☐ A carbon dioxide.
- ☐ B chlorophyll.
- ☐ C a chlorofluorocarbon.
- ☐ D cochineal.

- 12** Tubes carrying water and dissolved mineral salts from the roots to the leaves. This describes:

- ☐ A phloem.
- ☐ B plankton.
- ☐ C chloroplast.
- ☐ D xylem.

- 13** Which one of the following is found in plant cells but *not* in animal cells?

- ☐ A A nucleus
- ☐ B A cell wall
- ☐ C A cell membrane
- ☐ D Cytoplasm

- 14** Flowers are pollinated by:

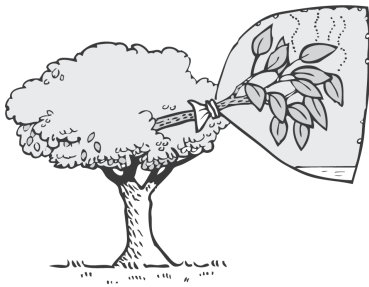
- ☐ A the wind.
- ☐ B insects.
- ☐ C birds.
- ☐ D all of the above.

- 15** Tubes carrying sugars from the leaves to other parts of the plant.

This refers to:

- ☐ A pigments.
- ☐ B petals.
- ☐ C phloem.
- ☐ D proteins.

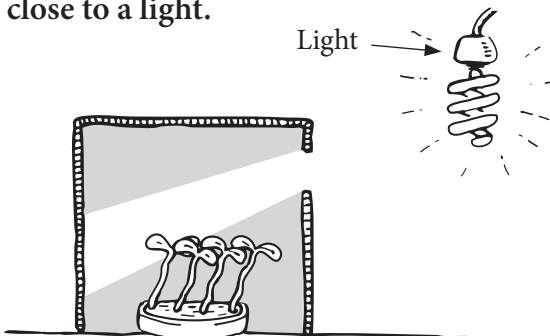
- 16** A student put a plastic bag over some leaves on a branch of a tree during sunny weather, as shown in the illustration.



After a few days, the bag had some water in the bottom of it. This is due to a process called:

- ☐ A perspiration.
- ☐ B respiration.
- ☐ C expiration.
- ☐ D transpiration.

- 17** A student grew some seeds in a Petri dish lid. The resulting seedlings were placed in a box which had a hole in the side, and left close to a light.



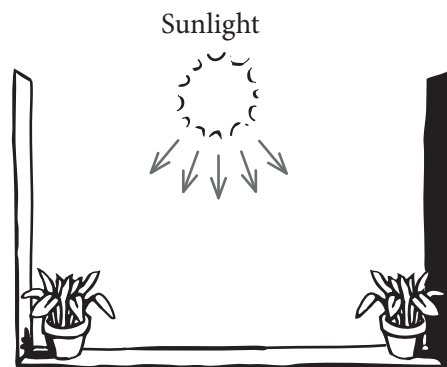
After a few days, it was found that the seedlings were no longer growing straight up, but were growing towards the light. This is called:

- ☐ A the greenhouse effect.
- ☐ B phototropism.
- ☐ C germination.
- ☐ D geotropism.

- 18** Dicotyledons are plants that:

- ☐ A flower every two years.
- ☐ B only last for two years after planting.
- ☐ C develop two leaves when germinating.
- ☐ D can be either male or female.

- 19** The label attached to an indoor plant states: *This plant will grow faster if placed near white or pastel-coloured walls than if placed next to dark coloured walls.* A student set up the experiment as shown to test this statement.



To ensure that the results obtained were valid (correct), the student would also have to:

- ☐ A use different types of indoor plants.
- ☐ B use different sorts of outdoor plants.
- ☐ C use more than one plant of the same type.
- ☐ D use different room temperatures.

- 20** The diagram, which is drawn to scale, shows a TV presenter who is two metres tall, describing a tree. What is the height of this tree?

- ☐ A 4.6 m
- ☐ B 6.7 m
- ☐ C 7.6 m
- ☐ D 8.0 m



1 Which one of the following helps the body to resist infection by micro-organisms?

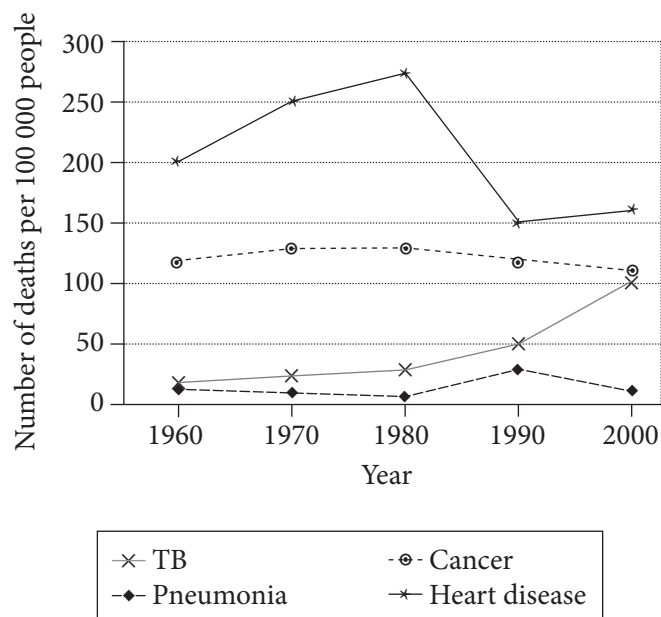
- ☐ A The immune system
- ☐ B Red blood cells
- ☐ C Haemoglobin
- ☐ D Genes

2 A disease that can be caught from another living creature is described as:

- ☐ A congenital.
- ☐ B infectious.
- ☐ C chronic.
- ☐ D widespread.

Questions 3–4 refer to the information below.

The line graph below shows the rates of death per 100 000 people due to four major diseases, in a specific country with a total population of 25 million people.



3 Which disease had the least change in number of deaths in the period shown?

- ☐ A TB
- ☐ B Pneumonia
- ☐ C Cancer
- ☐ D Heart disease

4 What was the total number of people who died of cancer in 1990?

- ☐ A 120
- ☐ B 150
- ☐ C 3 000
- ☐ D 30 000

5 Cystic fibrosis, haemophilia, muscular dystrophy and colour blindness are all:

- ☐ A inevitably fatal.
- ☐ B inherited.
- ☐ C caused by bacteria.
- ☐ D caused by viruses.

6 A pathogen:

- ☐ A cures a disease.
- ☐ B prevents a disease.
- ☐ C causes a disease.
- ☐ D is a sick person.

7 Practising cleanliness to assist the maintenance of health.

This is a definition of:

- ☐ A medicine.
- ☐ B detergent.
- ☐ C medication.
- ☐ D hygiene.

8 A researcher was investigating the occurrence of facial tumours in Tasmanian Devils. She was studying two groups of these animals. After doing some research, she decided that a particular virus was responsible for these tumours. In order to best test her hypothesis, she should:

- ☐ A inject both groups with the virus.
- ☐ B leave both groups to see if they contract the disease naturally.
- ☐ C inject one group with the virus and the other group with a different virus.
- ☐ D inject one group with the virus and leave the other group virus-free.

9 Leukaemia, rickets, arthritis and spina bifida are all diseases associated with the:

- ☐ A heart.
- ☐ B lungs.
- ☐ C skeleton.
- ☐ D brain.

10 Antibiotics:

- ☐ A cure all diseases.
- ☐ B were invented over 100 years ago.
- ☐ C only cure diseases caused by viruses.
- ☐ D only cure diseases caused by bacteria.

11 Microbes:

- ☐ A are always harmful to human beings.
- ☐ B are always helpful to human beings.
- ☐ C can be seen by the naked eye.
- ☐ D can only be seen with the aid of a microscope.

12 Which one of the following is an example of a non-infectious disease?

- ☐ A Lung cancer
- ☐ B Ross River fever
- ☐ C Influenza
- ☐ D Smallpox

13 *An ultramicroscopic infectious agent that replicates itself only within cells of living hosts; many of which can cause disease.*

What does this define?

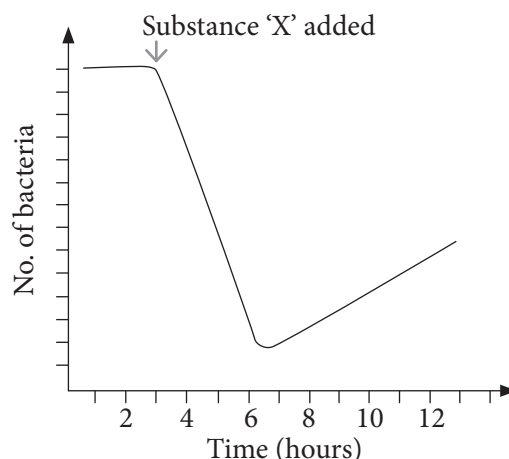
- ☐ A Virus
- ☐ B Bacterium
- ☐ C Fungus
- ☐ D Amoeba

14 A 'chronic' disease is one that:

- ☐ A always kills the patient.
- ☐ B lasts a long time.
- ☐ C has severe symptoms.
- ☐ D is very contagious.

Questions 15–16 refer to the information below.

The line graph below shows how the numbers of a pure culture of bacteria change after the addition of a substance 'X'.



15 Substance 'X' is most likely to be:

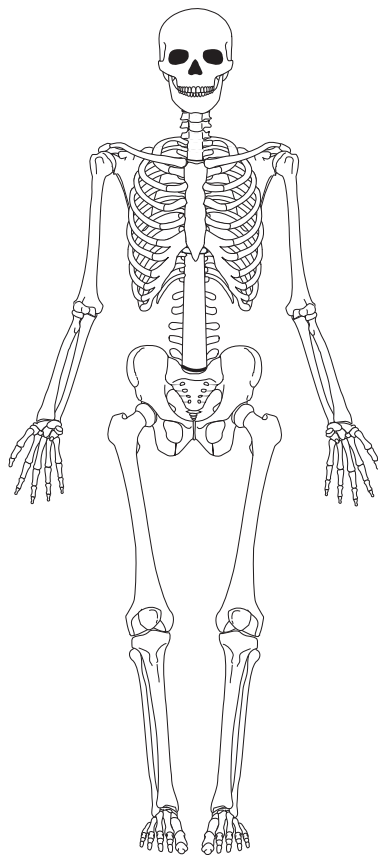
- ☐ A more of the same type of bacteria.
- ☐ B food.
- ☐ C an antibiotic.
- ☐ D water.

16 How long after the addition of 'X' did it take for the total number of bacteria to fall to half of their initial number?

- ☐ A 2 hours
- ☐ B 3 hours
- ☐ C 5 hours
- ☐ D 7 hours

17 The disease malaria is endemic in many parts of Africa. This means that in those areas, the disease is:

- ☐ A very common.
- ☐ B very rare.
- ☐ C more infectious than in other countries.
- ☐ D less infectious than in other countries.



- 1** In all mammals, the skeletal system:
- ☐ A provides protection for internal organs.
 - ☐ B is part of the endocrine system.
 - ☐ C has the same number of bones for each animal.
 - ☐ D is inflexible.

- 2** Which is the most important substance needed for bones to develop properly?

- ☐ A Carbohydrates
- ☐ B Fats
- ☐ C Calcium
- ☐ D Fluoride

- 3** Which parts of the body act as simple levers?

- ☐ A Bones
- ☐ B Muscles
- ☐ C Ligaments
- ☐ D Tendons

- 4** 'Ball & socket', 'hinged' and 'fixed' are all types of:

- ☐ A muscles.
- ☐ B bones.
- ☐ C joints.
- ☐ D nerves.

- 5** How many bones are there in the adult human skeleton?

- ☐ A 26
- ☐ B 62
- ☐ C 206
- ☐ D 602

- 6** The cranium refers to which area of the skeleton?

- ☐ A The head
- ☐ B The chest
- ☐ C The pelvis
- ☐ D The legs

- 7** Like most parts of the body, bones have an 'everyday' name as well as a scientific name. Which row of the table correctly links the everyday and scientific names for each bone?

	Clavicle	Femur	Patella	Scapula
A	backbone	thigh bone	shin bone	chest bone
B	spinal cord	knee cap	collar bone	shoulder blade
C	shoulder blade	knee cap	thigh bone	collar bone
D	collar bone	thigh bone	knee cap	shoulder blade

- ☐ A ☐ B ☐ C ☐ D

8 Canines, molars and incisors are all types of:

- ☐ A teeth.
- ☐ B joints.
- ☐ C muscles.
- ☐ D bone diseases.

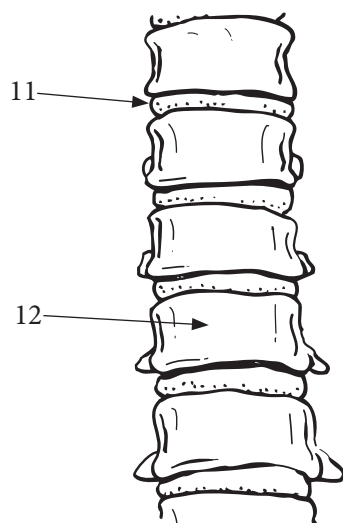
9 Bones are connected together by:

- ☐ A tendons.
- ☐ B cartilage.
- ☐ C ligaments.
- ☐ D synovials.

10 Muscles are attached to bones by:

- ☐ A tendons.
- ☐ B cartilage.
- ☐ C ligaments.
- ☐ D synovials.

Questions 11–12 refer to the diagram below of a section of the human backbone.



11 The part labelled '11' is:

- ☐ A synovial fluid.
- ☐ B a cartilage disc.
- ☐ C a vertebra.
- ☐ D a nerve cell.

12 The part labelled '12' is the:

- ☐ A synovial fluid.
- ☐ B a cartilage disc.
- ☐ C a vertebra.
- ☐ D a nerve cell.

13 The radius and ulna are bones located in the:

- ☐ A legs.
- ☐ B neck.
- ☐ C arms.
- ☐ D feet.

14 Humans can be described as 'vertebrates' because they:

- ☐ A have a spinal cord.
- ☐ B stand upright.
- ☐ C have joints between bones.
- ☐ D have bones that can flex.

15 Muscles always act:

- ☐ A singly.
- ☐ B in pairs.
- ☐ C in threes.
- ☐ D only when we think about them.

16 Another name for cartilage is:

- ☐ A brain cells.
- ☐ B spinal fluid.
- ☐ C bone.
- ☐ D gristle.

17 The front part of the ribcage is known as the:

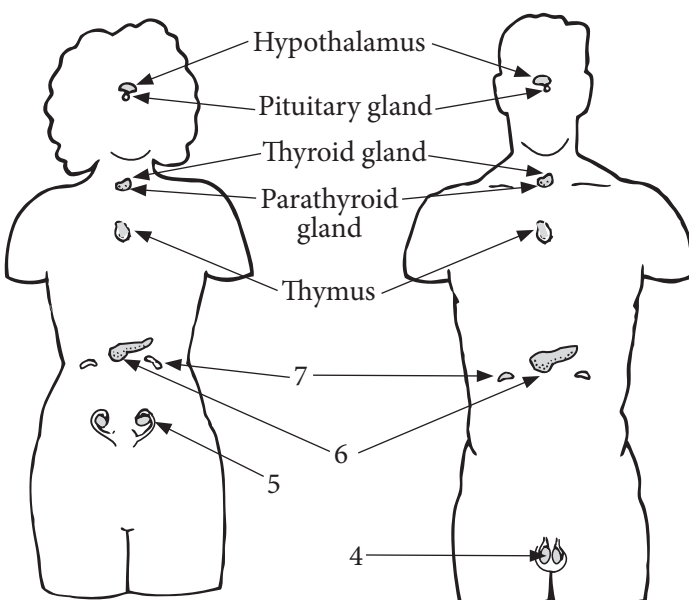
- ☐ A sternum.
- ☐ B ossicles.
- ☐ C suture.
- ☐ D thorax.

- 1** The endocrine system:
- ☐ A uses electrical impulses through nerves to make effects.
 - ☐ B works by releasing hormones.
 - ☐ C is located in the brain.
 - ☐ D controls all of the body's responses to stimuli.
- 2** Hormones:
- ☐ A are produced by glands.
 - ☐ B travel in the bloodstream.
 - ☐ C are chemicals.
 - ☐ D are all of the above.
- 3** The principal controller of hormone production is the:
- ☐ A liver.
 - ☐ B gonads.
 - ☐ C hypothalamus.
 - ☐ D pancreas.

- 4** The part labelled '4' is the:
- ☐ A adrenal gland.
 - ☐ B ovary.
 - ☐ C pancreas.
 - ☐ D testis.
- 5** The part labelled '5' is the:
- ☐ A adrenal gland.
 - ☐ B ovary.
 - ☐ C pancreas.
 - ☐ D testis.
- 6** The part labelled '6' is the:
- ☐ A adrenal gland.
 - ☐ B ovary.
 - ☐ C pancreas.
 - ☐ D testis.
- 7** The part labelled '7' is the:
- ☐ A adrenal gland.
 - ☐ B ovary.
 - ☐ C pancreas.
 - ☐ D testis.

- 8** The hormone that controls levels of sugar in the blood is:
- ☐ A adrenaline.
 - ☐ B insulin.
 - ☐ C oestrogen.
 - ☐ D thyroxine.
- 9** The hormone responsible for the 'fright, fight or flight' response is:
- ☐ A adrenaline.
 - ☐ B insulin.
 - ☐ C oestrogen.
 - ☐ D testosterone.

Questions 4–7 refer to the diagram below of some glands in the human body.



110 Sex hormones are produced by the:

- ☐ A adrenal gland.
- ☐ B gonads.
- ☐ C thyroid gland.
- ☐ D thymus.

111 Which gland is sometimes called the 'master gland' because it controls most of the other glands?

- ☐ A Adrenal
- ☐ B Parathyroid
- ☐ C Pituitary
- ☐ D Thyroid

112 The thyroid:

- ☐ A regulates the rate of the metabolism.
- ☐ B affects the rate of growth of the human body.
- ☐ C controls the sensitivity of the body's other hormones.
- ☐ D does all of the above.

113 Follicle-stimulating hormone (FSH) is:

- ☐ A produced only by men.
- ☐ B produced only by women.
- ☐ C produced by both men and women.
- ☐ D not produced by humans.

114 *Chemical 'messengers' which control various activities inside an organism.*
What does this define?

- ☐ A proteins
- ☐ B enzymes
- ☐ C hormones
- ☐ D antibodies

115 Hormones differ from nerve impulses in that hormones:

- ☐ A travel along nerve cells.
- ☐ B cause slower reactions than nerve cells.
- ☐ C cause quicker reactions than nerve impulses.
- ☐ D are electrical impulses.

116 Areas in which hormones exert their effects are:

- ☐ A reproduction, growth and development.
- ☐ B control of the internal environment.
- ☐ C regulation of energy production.
- ☐ D all of the above.

117 The level of glucose in blood is monitored by the pancreas. If the glucose level becomes too high, the pancreas produces hormones to reduce it. This is an example of:

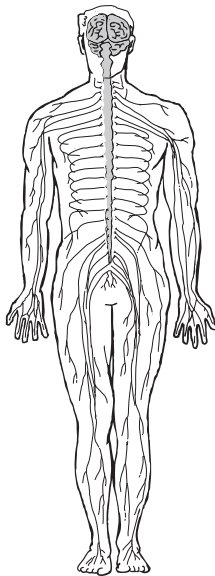
- ☐ A respiration.
- ☐ B a feedback mechanism.
- ☐ C digestion.
- ☐ D autotrophism.

118 The coordination of the various organ systems in humans is controlled by:

- ☐ A the nervous system only.
- ☐ B the endocrine system only.
- ☐ C the nervous system and the endocrine system.
- ☐ D hormones only.

119 The endocrine system assists in homeostasis. Homeostasis:

- ☐ A keeps conditions in the body the same.
- ☐ B keeps conditions in the body relatively stable.
- ☐ C only occurs during puberty.
- ☐ D helps us with our sense of balance.

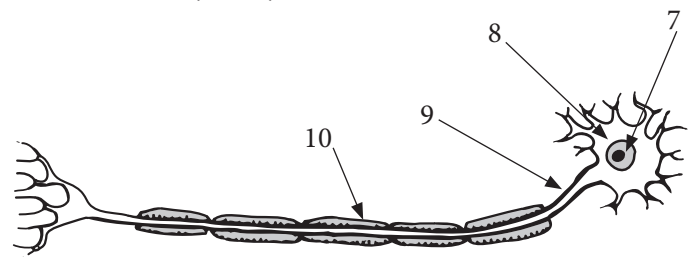


- 1** The body is controlled by the endocrine system and:
 - A** the autonomic system.
 - B** the nervous system.
 - C** neurotransmitters.
 - D** stimuli.
- 2** The central nervous system is made up of:
 - A** the brain and the spinal cord.
 - B** the brain.
 - C** neurones.
 - D** glands and nerve cells.
- 3** The medulla, cerebrum and cerebellum are areas of the:
 - A** spinal cord.
 - B** brain.
 - C** ear.
 - D** eye.
- 4** Anything that provokes a response from the body is called:
 - A** an impulse.
 - B** a response.
 - C** a stimulus.
 - D** sensation.

- 5** What is the missing word?
The nervous system is concerned with the reception of stimuli and the _____ of nerve impulses.
 - A** Reception
 - B** Interception
 - C** Translation
 - D** Transmission
- 6** Neurones are:
 - A** nerve cells.
 - B** types of hormones.
 - C** cells with no nuclei.
 - D** only found in the brain.

Questions 7–10 refer to the information below.

The diagram shows the parts of a neurone. This cell is composed of a cell body containing a nucleus and a long, thin fibre called an axon, along which an electrical signal travels. The axon is insulated by a myelin sheath.



- 7** The part labelled '7' is the:
 - A** axon.
 - B** cell body.
 - C** myelin sheath.
 - D** nucleus.
- 8** The part labelled '8' is the:
 - A** axon.
 - B** cell body.
 - C** myelin sheath.
 - D** nucleus.

9 The part labelled '9' is the:

- ☐ A axon.
- ☐ B cell body.
- ☐ C myelin sheath.
- ☐ D nucleus.

10 The part labelled '10' is the:

- ☐ A axon.
- ☐ B cell body.
- ☐ C myelin sheath.
- ☐ D nucleus.

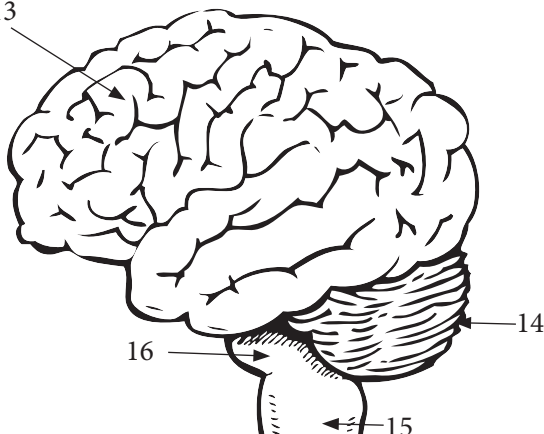
11 The junction between neurones is called:

- ☐ A a synapse.
- ☐ B a dendrite.
- ☐ C an effector.
- ☐ D a connection.

12 Which of the following shows the correct sequence for the body responding to an outside influence?

- ☐ A Sensory neurone → receptor cell → stimulus → motor neurone
- ☐ B Stimulus → receptor cell → sensory neurone → motor neurone
- ☐ C Stimulus → sensory neurone → receptor cell → motor neurone
- ☐ D Motor neurone → sensory neurone → receptor cell → stimulus

Questions 13–16 refer to the diagram below of the brain.



13 The part labelled '13' is the:

- ☐ A cerebellum.
- ☐ B frontal lobe.
- ☐ C medulla.
- ☐ D spinal cord.

14 The part labelled '14' is the:

- ☐ A cerebellum.
- ☐ B frontal lobe.
- ☐ C medulla.
- ☐ D spinal cord.

15 The part labelled '15' is the:

- ☐ A cerebellum.
- ☐ B frontal lobe.
- ☐ C medulla.
- ☐ D spinal cord.

16 The part labelled '16' is the:

- ☐ A cerebellum.
- ☐ B frontal lobe.
- ☐ C medulla.
- ☐ D spinal cord.

17 Which one of the following maintains homeostasis in the body?

- ☐ A The autonomic nervous system
- ☐ B The automatic nervous system
- ☐ C The cerebrum
- ☐ D The thalamus

18 Sometimes a message from a nerve (e.g. pain) doesn't go to the brain, but bypasses it and returns via the spinal cord. This is:

- ☐ A an instinct.
- ☐ B a slow process.
- ☐ C a reflex action.
- ☐ D a conscious reaction.

- 1** Which statement about the heart is *not* true?
- ☐ A The heart is a specialised muscle.
 - ☐ B The heart controls our feelings.
 - ☐ C The heart is connected to blood vessels.
 - ☐ D The heart pumps blood around the body.

- 2** When healthy individuals exercise and then rest, their pulse rate will:

- ☐ A increase and stay high.
- ☐ B increase and then return to normal.
- ☐ C decrease and then return to normal.
- ☐ D not be affected.

- 3** Red cells, white cells, plasma and platelets are all constituents of:

- ☐ A urine.
- ☐ B sweat.
- ☐ C exhaled air.
- ☐ D blood.

- 4** Which one of the following is used to listen to a heartbeat?

- ☐ A A microscope
- ☐ B A stethoscope
- ☐ C A sphygmomanometer
- ☐ D A periscope

- 5** How many chambers are there in the human heart?

- ☐ A 1
- ☐ B 2
- ☐ C 3
- ☐ D 4

- 6** Which fluid in the body contains special cells to kill germs?

- ☐ A Urine
- ☐ B Blood
- ☐ C Saliva
- ☐ D Perspiration

- 7** Blood is prevented from moving in the wrong direction by:

- ☐ A arteries.
- ☐ B valves.
- ☐ C veins.
- ☐ D capillaries.

- 8** Blood in arteries:

- ☐ A travels away from the heart.
- ☐ B travels towards the heart.
- ☐ C doesn't move.
- ☐ D doesn't have a pulse.

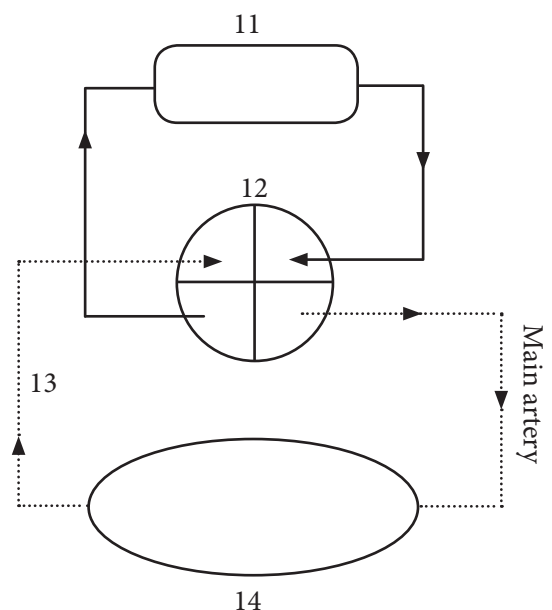
- 9** The term 'cardiac' refers to:

- ☐ A blood.
- ☐ B blood vessels.
- ☐ C the heart.
- ☐ D the brain.

- 10** Which one of the following shows a correct sequence of the flow of blood?

- ☐ A Capillaries, arteries, veins, heart
- ☐ B Capillaries, heart, veins, arteries
- ☐ C Heart, arteries, veins, capillaries
- ☐ D Heart, arteries, capillaries, veins

Questions 11–14 refer to the diagram below which shows a simplified version of the circulatory system.



11 The part labelled '11' is the:

- ☐ A body.
- ☐ B heart.
- ☐ C lungs.
- ☐ D main vein.

12 The part labelled '12' is the:

- ☐ A body.
- ☐ B heart.
- ☐ C lungs.
- ☐ D main vein.

13 The part labelled '13' is the:

- ☐ A body.
- ☐ B heart.
- ☐ C lungs.
- ☐ D main vein.

14 The part labelled '14' is the:

- ☐ A body.
- ☐ B heart.
- ☐ C lungs.
- ☐ D main vein.

15 The heart pumps blood by:

- ☐ A contracting.
- ☐ B expanding.
- ☐ C relaxing.
- ☐ D lengthening.

16 The 'beats' of a heart are:

- ☐ A controlled by the level of carbon dioxide in the blood.
- ☐ B controlled by the brain.
- ☐ C controlled by nerve impulses produced inside the heart.
- ☐ D dependent on the body's temperature.

17 Which row of the table correctly shows the features of the organs?

	Artery	Capillary	Vein
A	has valves	very fine tube	returns blood to heart
B	takes blood from heart	connects artery to vein	has valves
C	returns blood to heart	very fine tube	has thick wall
D	has thick wall	has valves	returns blood to heart

- ☐ A A ☐ B B ☐ C C ☐ D D

18 What is the role of platelets?

- ☐ A They make antibodies.
- ☐ B They carry oxygen.
- ☐ C They carry food.
- ☐ D They cause blood to clot.

19 Which statement about red blood cells is true?

- ☐ A They don't have a nucleus.
- ☐ B They are made in bone marrow.
- ☐ C They contain haemoglobin.
- ☐ D All of the above.

20 Which statement about white blood cells is *not* true?

- ☐ A They are the same shape as red blood cells.
- ☐ B They are fewer in number than red blood cells.
- ☐ C Some can digest germs and dead cells.
- ☐ D Some can produce antibodies.

21 The upper chambers of the heart are called:

- ☐ A atria.
- ☐ B atrials.
- ☐ C aortas.
- ☐ D ventricles.

- 1** In humans, the process of respiration involves:
- ☐ A breathing.
 - ☐ B external respiration.
 - ☐ C internal respiration.
 - ☐ D all of the above.
- 2** Which of the following shows the correct sequence of air entering the human body?
- ☐ A Mouth, trachea, bronchi, bronchioles, alveoli, capillaries
 - ☐ B Mouth, alveoli, bronchi, bronchioles, trachea, capillaries
 - ☐ C Nose, alveoli, bronchi, bronchioles, trachea, capillaries
 - ☐ D Capillaries, alveoli, bronchioles, bronchi, trachea, mouth

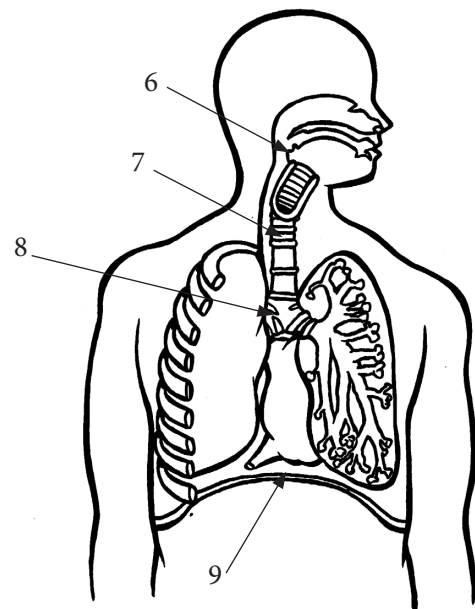
- 3** Which row of the table correctly matches each part of the respiratory system with its description?

	Alveolus	Bronchiole	Capillary	Trachea
A	plural is 'alveoli'	there are two of these	minute air sacs	also called the gullet
B	there are two of these	tiny blood vessels	minute air sacs	connects nose and mouth to the lungs
C	minute air sacs	small tubes	tiny blood vessels	also called the wind pipe
D	also called the wind pipe	minute air sacs	tiny blood vessels	there are two of these

- ☐ A A ☐ B B ☐ C C ☐ D D

- 4** Areas of the respiratory tract have a moist and sticky lining that catches foreign particles. This substance is called:
- ☐ A cilia.
 - ☐ B mucus.
 - ☐ C allergens.
 - ☐ D antibodies.
- 5** When we inhale, which of the following *does not* occur?
- ☐ A The diaphragm contracts.
 - ☐ B The chest cavity expands.
 - ☐ C The ribcage lifts.
 - ☐ D Air is forced out of the lungs.

Questions 6–9 refer to the diagram below which shows a cross-section of the respiratory system.



- 6** The part labelled '6' is the:
- ☐ A bronchus.
 - ☐ B diaphragm.
 - ☐ C epiglottis.
 - ☐ D trachea.
- 7** The part labelled '7' is the:
- ☐ A bronchus.
 - ☐ B diaphragm.
 - ☐ C epiglottis.
 - ☐ D trachea.

- 8** The part labelled '8' is the:
- A** bronchus. **B** diaphragm.
C epiglottis. **D** trachea.

- 9** The part labelled '9' is the:
- A** bronchus. **B** diaphragm.
C epiglottis. **D** trachea.

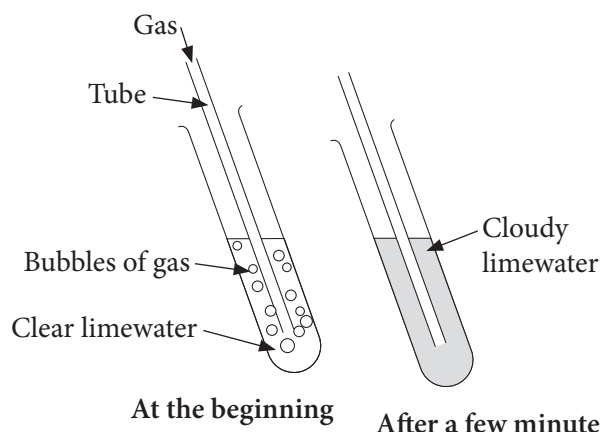
- 10** Which is the most accurate description of the lungs?

- A** They are two spongy bags of tissue.
B They are the only organs involved in respiration.
C They are two large, hollow sacs.
D They are mainly composed of muscle.

- 11** An airtight space enclosing the lungs. This best describes the:

- A** intercostal muscles.
B chest.
C pleural cavity.
D thorax.

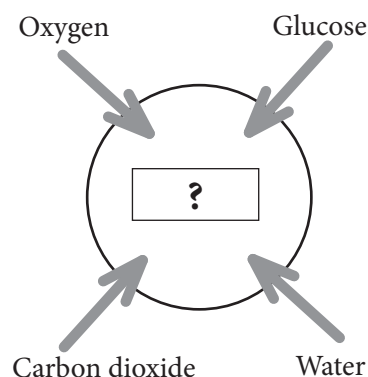
- 12** The diagram below shows what happens when a particular gas is passed through limewater for a few minutes.



Which statement best reflects what is shown in the diagram?

- A** The gas is nitrogen.
B The gas contains carbon dioxide.
C The gas is breathed out by humans.
D The gas contains oxygen.

- 13** The diagram represents respiration taking place in an animal cell.



What is the missing term?

- A** Nitrogen **B** Nerve impulses
C Waste **D** Energy

- 14** In the lungs, alveoli are surrounded by capillaries. This is the place where:

- A** gases are exchanged.
B food is transferred into cells.
C the balance of water is stabilised.
D harmful substances are filtered out.

- 15** The table below shows the relative composition of gases in the inhaled air and exhaled air of a typical person.

	Approximate composition of gas breathed in (%)	Approximate composition of gas breathed out (%)
Carbon dioxide	0.04	5.0
Oxygen	21	16
Nitrogen	78	78
Other gases	1.0	1.0

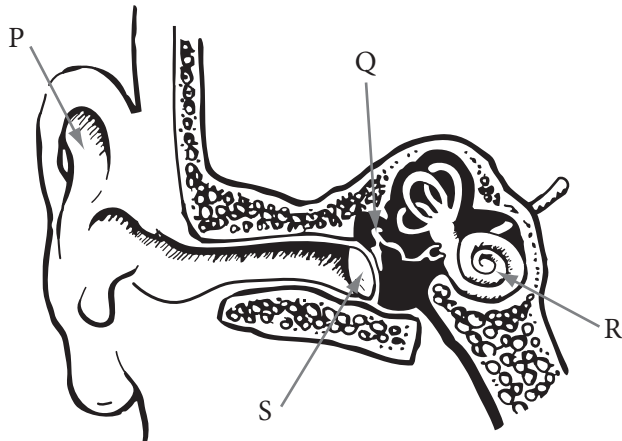
Based on the information in the table:

- A** humans take in more carbon dioxide than they breathe out.
B humans breathe out more carbon dioxide than they take in.
C oxygen is needed for respiration to take place.
D the amounts of gases in the atmosphere change from place to place.

1 Auditory refers to:

- ☐ A sight.
- ☐ B hearing.
- ☐ C touch.
- ☐ D thought.

Questions 2–5 refer to the diagram below which shows a section of the human hearing system.



2 The part labelled 'P' is the:

- ☐ A cochlea.
- ☐ B eardrum.
- ☐ C ossicles.
- ☐ D pinna.

3 The part labelled 'Q' is the:

- ☐ A cochlea.
- ☐ B eardrum.
- ☐ C ossicles.
- ☐ D pinna.

4 The part labelled 'R' is the:

- ☐ A cochlea.
- ☐ B eardrum.
- ☐ C ossicles.
- ☐ D pinna.

5 The part labelled 'S' is the:

- ☐ A cochlea.
- ☐ B eardrum.
- ☐ C ossicles.
- ☐ D pinna.

6 In which row of the table is the correct description matched with each part of the ear?

	Cochlea	Eardrum	Ossicles	Pinna
A	spiral shape	outer flap of skin	fluid-filled tube	small bones
B	fluid-filled tube	taut membrane	small bones	outer flap of skin
C	spiral shape	taut membrane	fluid-filled tube	outer flap of skin
D	small bones	outer flap of skin	spiral shape	taut membrane

- ☐ A A ☐ B B ☐ C C ☐ D D

7 In which row of the table is the correct function matched with each part of the ear?

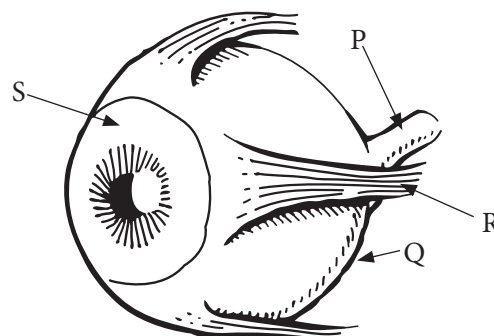
	Cochlea	Eardrum	Ossicles	Pinna
A	changes sound to electrical signals	vibrates when hit by sound waves	magnifies movement	directs sound into the ear
B	vibrates when hit by sound waves	magnifies movement	directs sound into the ear	changes sound to electrical signals
C	magnifies movement	directs sound into the ear	changes sound to electrical signals	helps with sense of balance
D	helps with sense of balance	vibrates when hit by sound waves	changes sound to electrical signals	directs sound into the ear

- ☐ A A ☐ B B ☐ C C ☐ D D

- 8** The upper (higher) range of human hearing is approximately:
- ☐ A 20 Hz.
 - ☐ B 200 Hz.
 - ☐ C 2000 Hz.
 - ☐ D 20 000 Hz.
- 9** The lower range of human hearing is approximately:
- ☐ A 20 Hz.
 - ☐ B 200 Hz.
 - ☐ C 2000 Hz.
 - ☐ D 20 000 Hz.
- 10** As people grow older:
- ☐ A their hearing becomes poorer.
 - ☐ B their hearing improves.
 - ☐ C their hearing remains unchanged.
 - ☐ D they become more sensitive to noise.
- 11** Sound above the usual range of human hearing is called:
- ☐ A infrasound.
 - ☐ B ultrasound.
 - ☐ C sonic.
 - ☐ D audio.
- 12** We measure how loud a sound is in:
- ☐ A watts.
 - ☐ B hertz.
 - ☐ C decibels.
 - ☐ D ohms.

- 13** The ear sends signals to the brain in the form of:
- ☐ A pulses of liquid.
 - ☐ B pulses of air.
 - ☐ C movements of bones.
 - ☐ D electrical impulses.

Questions 14–17 refer to the diagram below which shows the front of a human eye.



- 14** The part labelled 'P' is the:
- ☐ A iris.
 - ☐ B muscle.
 - ☐ C optic nerve.
 - ☐ D sclerotic coat.
- 15** The part labelled 'Q' is the:
- ☐ A iris.
 - ☐ B muscle.
 - ☐ C optic nerve.
 - ☐ D sclerotic coat.
- 16** The part labelled 'R' is the:
- ☐ A iris.
 - ☐ B muscle.
 - ☐ C optic nerve.
 - ☐ D sclerotic coat.

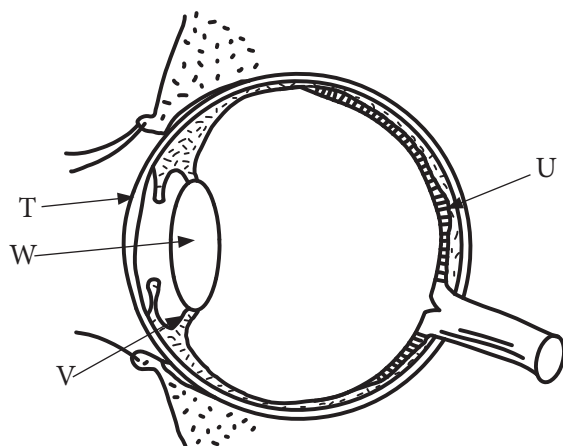
17 The part labelled 'S' is the:

- ☐ A iris.
- ☐ B muscle.
- ☐ C optic nerve.
- ☐ D sclerotic coat.

21 The part labelled 'W' is the:

- ☐ A cornea.
- ☐ B focusing muscle.
- ☐ C lens.
- ☐ D retina.

Questions 18–21 refer to the diagram below which shows a section of a human eye.



18 The part labelled 'T' is the:

- ☐ A cornea.
- ☐ B focusing muscle.
- ☐ C lens.
- ☐ D retina.

19 The part labelled 'U' is the:

- ☐ A cornea.
- ☐ B focusing muscle.
- ☐ C lens.
- ☐ D retina.

20 The part labelled 'V' is the:

- ☐ A cornea.
- ☐ B focusing muscle.
- ☐ C lens.
- ☐ D retina.

22 In which row of the table is the correct description matched with each part of the eye?

	Iris	External muscle	Optic nerve	Sclerotic coat
A	coloured	work in pairs	links retina to brain	transparent
B	coloured	work in pairs	links retina to brain	white
C	transparent	work in pairs	links retina to brain	white
D	circular	attached to sclerotic coat	made up of rods and cones	made up of a fluid

- ☐ A A ☐ B B ☐ C C ☐ D D

23 In which row of the table is the correct function matched with each part of the eye?

	Cornea	Focusing muscles	Lens	Retina
A	limits light coming into eye	changes shape of lens	focuses light	changes light to electrical impulses
B	refracts light	changes shape of lens	focuses light	changes light to electrical impulses
C	changes light to electrical impulses	moves eye up and down	limits light coming into eye	absorbs light
D	changes shape of lens	changes light to electrical impulses	absorbs light	limits light coming into eye

- ☐ A A ☐ B B ☐ C C ☐ D D

24 The brain receives information from many sources. Which percentage represents the amount of information the brain receives from our eyes?

- ☐ A 10%
- ☐ B 20%
- ☐ C 40%
- ☐ D 80%

25 The retina is made up of two types of photosensitive cells – rods and cones. Which row of the table shows the correct function of rods and cones?

	Rods	Cones
A	responsible for daylight vision	responsible for night vision
B	sensitive to colours	sensitive to light
C	responsible for night vision	sensitive to colours
D	sensitive to colours	sensitive to colours

- ☐ A A ☐ B B ☐ C C ☐ D D

26 Humans have binocular vision. This means that they:

- ☐ A have two eyes.
- ☐ B can judge distances.
- ☐ C have 3-dimensional vision.
- ☐ D have all of the above.

27 Colour blindness is a condition where people are unable to distinguish between colours. Of the population, 0.005% are totally colour blind. Assuming the population of Australia is 20 500 000, how many people have this condition?

- ☐ A 103
- ☐ B 1 025
- ☐ C 10 250
- ☐ D 102 500

28 What is the function of tears in the eyes?

- ☐ A Lubricate the eyeball
- ☐ B Prevent the membranes around the eye from drying out
- ☐ C Kill bacteria.
- ☐ D All of the above

29 When someone is described as being *myopic*, they:

- ☐ A can focus better on nearby objects.
- ☐ B can focus better on distant objects.
- ☐ C have better than average eyesight.
- ☐ D have difficulty distinguishing between colours.

30 An action that we do automatically, without thought, is best described as what type of action?

- ☐ A autonomic
- ☐ B thoughtless
- ☐ C mindless
- ☐ D reflex

31 The term 'tactile' refers to which sense?

- ☐ A sight
- ☐ B hearing
- ☐ C smell
- ☐ D touch

32 Which response correctly applies to the tongue and taste?

- ☐ A All people can taste equally well.
- ☐ B The tongue has chemoreceptors on its surface.
- ☐ C All areas of the tongue are equally sensitive to various flavours.
- ☐ D The basic four flavours are sweet, sour, acidic and bitter.

1 A single female sex cell is called:

- ☐ A a sperm.
- ☐ B an ova.
- ☐ C an ovum.
- ☐ D a zygote.

2 A single male sex cell is called:

- ☐ A a sperm.
- ☐ B an ova.
- ☐ C an ovum.
- ☐ D a zygote.

3 A gamete is:

- ☐ A always male.
- ☐ B always female.
- ☐ C either female or male.
- ☐ D multicellular.

4 Human sperm:

- ☐ A may carry the male chromosome.
- ☐ B contain no chromosomes.
- ☐ C are produced in small amounts.
- ☐ D are visible to the naked eye.

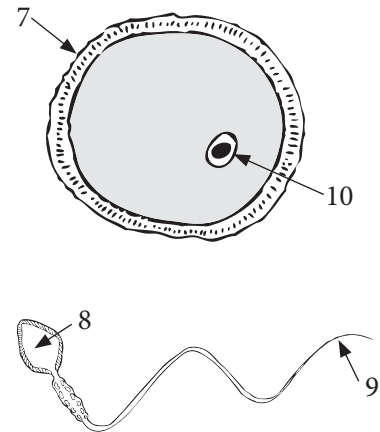
5 Female and male sex cells are produced by:

- ☐ A meiosis.
- ☐ B mitosis.
- ☐ C ovulation.
- ☐ D gestation.

6 In females and males, the reproductive system becomes active around the ages of 11 to 16 years. This time of life is called:

- ☐ A senility.
- ☐ B puberty.
- ☐ C youth.
- ☐ D conception.

Questions 7–12 refer to the diagram below which shows human sex cells.



7 The part labelled '7' is the:

- ☐ A flagellum.
- ☐ B gel coating.
- ☐ C head.
- ☐ D nucleus.

8 The part labelled '8' is the:

- ☐ A flagellum.
- ☐ B gel coating.
- ☐ C head.
- ☐ D nucleus.

9 The part labelled '9' is the:

- ☐ A flagellum.
- ☐ B gel coating.
- ☐ C head.
- ☐ D nucleus.

10 The part labelled '10' is the:

- ☐ A flagellum.
- ☐ B gel coating.
- ☐ C head.
- ☐ D nucleus.

11 In which one of the following do both parts contain chromosomes?

- ☐ A Flagellum and gel coating
- ☐ B Gel coating and head
- ☐ C Head and nucleus
- ☐ D Flagellum and nucleus

12 The diagrams shows the ovum magnified by $200\times$ and the sperm magnified by $2000\times$. If the sperm is 0.02 mm long, then the diameter of the ovum is nearest to:

- ☐ A 0.002 mm.
- ☐ B 0.02 mm.
- ☐ C 0.2 mm.
- ☐ D 2 mm.

13 Which row in the table shows the correct number of chromosomes present in a sperm and an ovum?

	Sperm	Ovum
A	23	23
B	23	46
C	46	23
D	46	46

- ☐ A A ☐ B B ☐ C C ☐ D D

14 Fertilisation means that:

- ☐ A hormone levels in the bloodstream have dropped.
- ☐ B an ova has been produced in an ovary.
- ☐ C a female sex cell and a male sex cell have joined.
- ☐ D a sex cell starts to divide.

15 A fertilised egg is called:

- ☐ A an ovum.
- ☐ B a zygote.
- ☐ C a foetus.
- ☐ D an embryo.

16 How many sperm will usually fertilise an ovum?

- ☐ A One
- ☐ B Two
- ☐ C Three
- ☐ D Four

17 Which one of the following shows the correct pathway of a human egg from its formation to the point of passing out of the female? Note that fertilisation has not taken place.

- ☐ A Uterus, oviduct, ovary, vagina
- ☐ B Ovary, uterus, ureter, bladder
- ☐ C Ovary, oviduct, uterus, vagina
- ☐ D Vagina, uterus, oviduct, ovary

18 When the uterus lining and the unfertilised egg break away and leave the body, this is called:

- ☐ A ovulation.
- ☐ B fertilisation.
- ☐ C menstruation.
- ☐ D gestation.

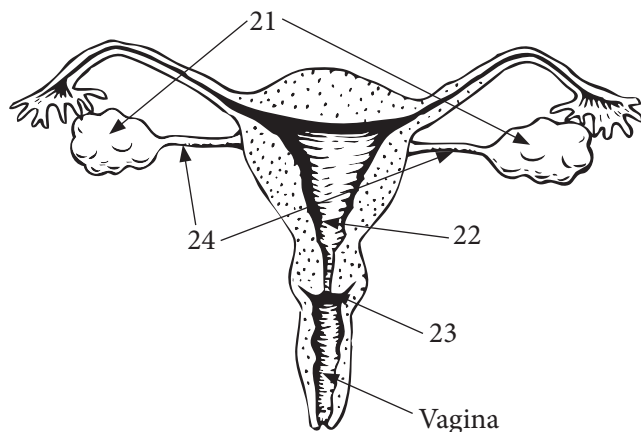
19 Gonads are:

- ☐ A cells.
- ☐ B hormones.
- ☐ C contraceptives.
- ☐ D sex organs.

20 Which type of contraception involves permanent sterilisation?

- ☐ A Condom
- ☐ B The 'pill'
- ☐ C Vasectomy
- ☐ D IUD

Questions 21–24 refer to the diagram below which shows the female reproductive system.



21 The part labelled '21' is the:

- ☐ A cervix.
- ☐ B ovary.
- ☐ C oviduct.
- ☐ D uterus.

22 The part labelled '22' is the:

- ☐ A cervix.
- ☐ B ovary.
- ☐ C oviduct.
- ☐ D uterus.

23 The part labelled '23' is the:

- ☐ A cervix.
- ☐ B ovary.
- ☐ C oviduct.
- ☐ D uterus.

24 The part labelled '24' is the:

- ☐ A cervix.
- ☐ B ovary.
- ☐ C oviduct.
- ☐ D uterus.

25 In which row of the table is each organ correctly matched to its feature?

Features				
	Cervix	Ovary	Oviduct	Uterus
A	also called the womb	produces sex hormones	also called fallopian tube	produces eggs
B	narrow outer end of uterus	produces eggs	tube through which eggs pass	also called the womb
C	wide inner end of uterus	there are two	produces eggs	produces sex hormones
D	produces eggs	produces sex hormones	there are two	where sperm meet ova

- ☐ A A ☐ B B ☐ C C ☐ D D

26 Human reproduction:

- ☐ A is sexual.
- ☐ B is asexual.
- ☐ C is controlled by the number of sperm released by a male.
- ☐ D results in exactly the same number of males and females being born.

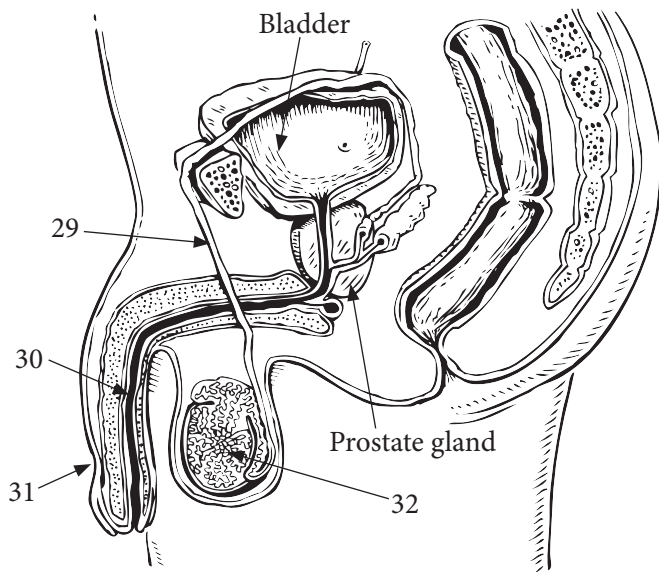
27 The fusion of gametes to produce a potential new organism of the same species. What does this define?

- ☐ A Cloning
- ☐ B Parthenogenesis
- ☐ C Fertilisation
- ☐ D Gastrulation

28 Two offspring resulting from the same pregnancy, either of the same or opposite sex. This best refers to:

- ☐ A progeny.
- ☐ B siblings.
- ☐ C twins.
- ☐ D mutations.

Questions 29–32 refer to the diagram below which shows the male reproductive system.



29 The part labelled '29' is the:

- ☐ A penis.
- ☐ B testis.
- ☐ C ureter.
- ☐ D vas deferens.

30 The part labelled '30' is the:

- ☐ A penis.
- ☐ B testis.
- ☐ C ureter.
- ☐ D vas deferens.

31 The part labelled '31' is the:

- ☐ A penis.
- ☐ B testis.
- ☐ C ureter.
- ☐ D vas deferens.

32 The part labelled '32' is the:

- ☐ A penis.
- ☐ B testis.
- ☐ C ureter.
- ☐ D vas deferens.

33 The epididymis:

- ☐ A is a storage vessel for sperm.
- ☐ B manufactures sperm.
- ☐ C manufactures male hormones.
- ☐ D is a gland.

34 Which statement about female ovulation and fertility is true?

- ☐ A Almost all sex cells never fully develop and are reabsorbed by the body.
- ☐ B Ova are continually being produced by the ovaries.
- ☐ C Almost all ova that are released are fertilised by sperm.
- ☐ D Menopause is the stage when women's ovulation increases.

35 The time for which a baby is in the uterus is called:

- ☐ A gestation.
- ☐ B puberty.
- ☐ C fertilisation.
- ☐ D development.

36 Which organ allows nutrients and oxygen to go to a developing foetus and takes away waste products?

- ☐ A Placenta
- ☐ B Amnion
- ☐ C Uterus
- ☐ D Navel

- 1** *The branch of biology that deals with the heredity and variation of organisms. This describes:*
- ☐ A anatomy. ☐ B genetics.
☐ C physiology. ☐ D phrenology.
- 2** *Who was the Austrian/Czech monk who first investigated inheritance in sweet peas?*
- ☐ A Russell ☐ B Wallace
☐ C Darwin ☐ D Mendel
- 3** *Gregor Mendel is frequently referred to as the 'father of genetics.' This is because he:*
- ☐ A discovered that genes act in pairs.
☐ B discovered DNA.
☐ C devised a theory predicting the results of crossing plants.
☐ D came up with the first theory regarding DNA.
- 4** *The genetic information of an individual is called its:*
- ☐ A phenotype. ☐ B genotype.
☐ C sex. ☐ D inheritance.
- 5** *Which of the following describes a gene?*
- ☐ A It is the functional unit of inheritance.
☐ B It is a specific sequence of nucleotides in DNA or RNA.
☐ C It causes the expression of one or more physical traits.
☐ D All of the above.
- 6** *They are singular pieces of DNA containing many genes and are found in the nucleus of cells. They are stained very strongly by some dyes. This describes:*
- ☐ A chromosomes. ☐ B hormones.
☐ C nucleotides. ☐ D nucleic acids.
- 7** *The genes that determine the characteristics of a human child come:*
- ☐ A from the male parent only.
☐ B from the female parent only.
☐ C from neither parent.
☐ D half from each parent.
- 8** *For many characteristics in organisms, there can be two alternative forms of the gene. These forms are called:*
- ☐ A genomes. ☐ B diploids.
☐ C alleles. ☐ D aberrations.
- 9** *Only one of the genes mentioned in question 8 can be expressed. This one is said to be:*
- ☐ A successful. ☐ B dominant.
☐ C leading. ☐ D recessive.
- 10** *The type of cell division which results in two cells identical to the parent cell is:*
- ☐ A meiosis. ☐ B mitosis.
☐ C mutation. ☐ D diffusion.
- 11** *A permanent change, a structural alteration, in the DNA or RNA of an organism. This describes:*
- ☐ A mutation. ☐ B evolution.
☐ C cancer. ☐ D extinction.
- 12** *The physical appearance of a particular characteristic is the:*
- ☐ A genotype. ☐ B phenotype.
☐ C gender. ☐ D pedigree.

13 With the exception of identical twins, all the people in the world will be genetically different from each other. This is because:

- ☐ A in their lifetime males make millions of sperm cells.
- ☐ B as we grow older, our cells constantly replace themselves.
- ☐ C there is a huge number of combinations of genes.
- ☐ D mitosis never replaces two cells genetically identical.

14 Which row of the table shows the correct arrangement of sex chromosomes for females and males?

	Female	Male
A	XY	XX
B	YY	XX
C	XX	YY
D	XX	XY

- ☐ A ☐ B ☐ C ☐ D

15 DNA chains contain adenine (A), cytosine (C), guanine (G) and thymine (T). These link together in a specific sequence, as shown in the left of the diagram.

	DNA chain	A	B	C	D
A-T	T-	T	A	A	A
G-C	A-	C	T	T	T
C-G	G-	G	G	C	G
T-A	T-	A	C	A	C
T-A	A-	A	T	T	T
A-T	A-	T	A	T	A
C-G	C-	G	C	G	C
G-C	G-	C	G	C	G

Which of the columns on the right of the diagram shows the correct sequence that would link with the incomplete DNA chain?

- ☐ A ☐ B ☐ C ☐ D

16 Which one of the following best describes the structure of DNA?

- ☐ A Single helix
- ☐ B Double helix
- ☐ C Triple helix
- ☐ D Ladder-like, with rungs

17 Which scientists are credited with the discovery of the structure of DNA?

- ☐ A Walter Sutton and Theodore Boveri
- ☐ B Lise Meitner and Otto Hahn
- ☐ C Pierre and Marie Curie
- ☐ D James Watson and Francis Crick

18 Sometimes in humans, one fertilised egg (ovum) splits and then develops into two babies with exactly the same genetic information. These babies are:

- ☐ A fraternal twins.
- ☐ B identical twins.
- ☐ C dizygotic twins.
- ☐ D much more likely to be male than female.

19 An organism that has had part of another species' genetic information transferred into its own is referred to as:

- ☐ A a genome.
- ☐ B transgenic.
- ☐ C a hybrid.
- ☐ D a clone.

20 Adenine, guanine, cytosine and thymine are all:

- ☐ A amino acids.
- ☐ B inorganic acids.
- ☐ C sugars.
- ☐ D proteins.

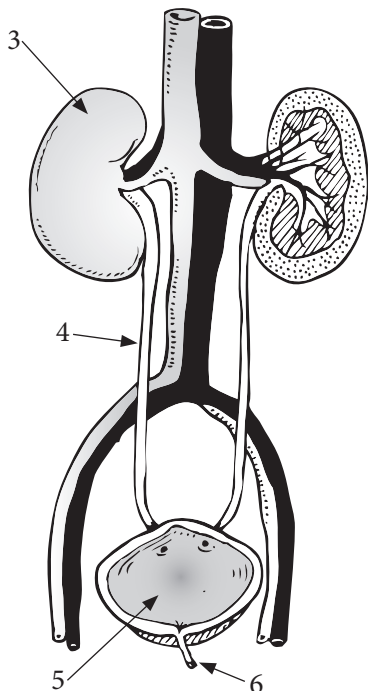
1 This organ adjusts the level of glucose in the blood, converts poisonous substances into less harmful ones and removes old red blood cells. It is the:

- ☐ A bladder.
- ☐ B gall bladder.
- ☐ C kidney.
- ☐ D liver.

2 This organ removes nitrogen-rich compounds and excess salts from the blood. It is the:

- ☐ A bladder.
- ☐ B gall bladder.
- ☐ C kidney.
- ☐ D liver.

Questions 3–6 refer to the diagram below.



3 The part labelled '3' is the:

- ☐ A bladder.
- ☐ B kidney.
- ☐ C ureter.
- ☐ D urethra.

4 The part labelled '4' is the:

- ☐ A bladder.
- ☐ B kidney.
- ☐ C ureter.
- ☐ D urethra.

5 The part labelled '5' is the:

- ☐ A bladder.
- ☐ B kidney.
- ☐ C ureter.
- ☐ D urethra.

6 The part labelled '6' is the:

- ☐ A bladder.
- ☐ B kidney.
- ☐ C ureter.
- ☐ D urethra.

7 The word *renal* refers to which organ?

- ☐ A Kidney
- ☐ B Heart
- ☐ C Liver
- ☐ D Lung

8 The word *hepatic* refers to which organ?

- ☐ A Kidney
- ☐ B Heart
- ☐ C Liver
- ☐ D Lung

9 The kidney is an organ:

- ☐ A of excretion.
- ☐ B that secretes hormones.
- ☐ C that makes red blood cells.
- ☐ D that stores glucose.

10 The medulla and cortex are parts of which organ?

- ☐ A Kidney
- ☐ B Heart
- ☐ C Liver
- ☐ D Lung

- 1** Traces of life on Earth first appeared approximately how many millions of years ago?
- A** 4
 - B** 35
 - C** 350
 - D** 3 500
- 2** Which statement about fossils is true?
- A** All fossils are millions of years old.
 - B** All organisms leave fossil traces.
 - C** Fossils prove evolution has occurred.
 - D** Fossils show that organisms have changed over time.
- 3** The Wollemi Pine was believed to have died out millions of years ago. It was rediscovered a few years ago. What is the name for an organism that was thought to no longer exist, but then living examples are found?
- A** Extinct
 - B** A mutation
 - C** A living fossil
 - D** A clone
- 4** Which statement about evolution is true?
- A** It has been proved by scientific observations.
 - B** It has been disproved by scientific observations.
 - C** It is a scientific theory based on evidence.
 - D** It is only believed in by a small minority of scientists.
- 5** Populations of similar organisms that become geographically isolated from each other:
- A** adopt independent evolutionary paths.
 - B** remain exactly the same.
 - C** usually become extinct.
 - D** have increased rates of mutation.
- 6** *Early developmental stages of all vertebrates are similar due to a common ancestry. This is a conclusion that could be drawn by studying which aspect of vertebrates?*
- A** Comparative embryology
 - B** Mating habits
 - C** Their fossils
 - D** How quickly offspring develop
- 7** Giraffes have long necks. According to Lamarck, the giraffe's long neck came about because:
- A** of variation and natural selection.
 - B** giraffes stretch their necks trying to eat higher vegetation.
 - C** of extreme mutations.
 - D** their predators also have long necks.
- 8** Modern evolutionary theory states that variations between organisms arise due to:
- A** mutations.
 - B** random segregation of chromosomes.
 - C** random recombination of chromosomes.
 - D** all of the above.

9 *The study of similarities and differences in the structure of organisms.*

This describes:

- ☐ A zoology.
- ☐ B biology.
- ☐ C comparative anatomy.
- ☐ D phylogeny.

10 Comparative anatomy, DNA analysis, embryology, fossils and geographic distribution are evidence for:

- ☐ A evolution.
- ☐ B the age of the Earth.
- ☐ C how life arose on Earth.
- ☐ D how land masses on Earth gradually moved.

11 Which statement is consistent with current scientific theories of evolution?

- ☐ A Organisms have not changed over long periods of time.
- ☐ B Organisms have changed over long periods of time from simple to complex.
- ☐ C Organisms have changed over long periods of time from complex to simple.
- ☐ D Organisms only change when mutations occur.

12 One theory of evolution has five major themes. These are:

- overproduction of offspring
- competition for resources
- variation within a species
- survival of the fittest
- inheritance of successful variations.

This is the theory associated with:

- ☐ A Karl von Linnaeus.
- ☐ B James Usher.
- ☐ C Charles Darwin.
- ☐ D Jean Baptiste Lamarck.

13 *If an organism acquires a characteristic by use or disuse of an organ, then this characteristic is passed on to the next generation.*

Who proposed this theory of evolution?

- ☐ A Richard Dawkins
- ☐ B Steven Jay Gould
- ☐ C Charles Darwin
- ☐ D Jean Baptiste Lamarck

14 Which evolutionary theory states that stability in organisms is often marked by instances of rapid change? ('Fits and starts')

- ☐ A Abiogenesis
- ☐ B Punctuated equilibrium
- ☐ C Creationism
- ☐ D Speciation

15 As well as many other animals, Charles Darwin studied the finches on the Galapagos Islands. He noticed that these finches had different beaks depending upon which island they lived. He attributed this to the:

- ☐ A diet of the birds.
- ☐ B cold climate.
- ☐ C migratory habits of the birds.
- ☐ D aggressive behaviour of male birds towards other male birds.

16 *Remains or traces of an organism found naturally preserved in the Earth's crust.*
This defines a:

- ☐ A species.
- ☐ B fossil.
- ☐ C mineral.
- ☐ D culture.

- 1** The study of the relationship of organisms with their physical and biological surroundings.

This is a definition of:

- ☐ A biology.
- ☐ B ecology.
- ☐ C biometrics.
- ☐ D geology.

- 2** In which row of the table is the correct description matched with each ecological term?

	Habitat	Community	Population
A	place where organisms live	different species of organisms interacting	collection of same species
B	also known as an 'ecozone'	collection of same species	different species of organisms interacting
C	interaction between different species	diversity of organisms living in a particular area.	number of organisms living in a particular area
D	how suitable an area is for life to exist	collection of same species	how many humans are present in a place

- 3** Tropical rainforests, grasslands and deserts are major areas of similar ecologies and are also known as:

- ☐ A biomes.
- ☐ B domains.
- ☐ C regions.
- ☐ D provinces.

- 4** In ecology, when we refer to *biomass* we mean:

- ☐ A how heavy an organism is.
- ☐ B dead plant and animal matter.
- ☐ C all plant and animal matter.
- ☐ D matter that can biodegrade.

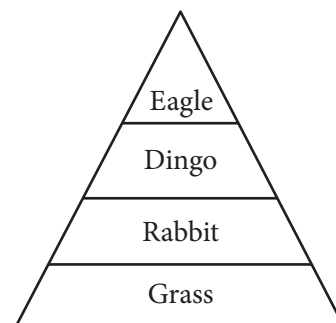
- 5** For an ecosystem to thrive, decomposers must be present. This is because they:

- ☐ A capture atmospheric nitrogen and change it to a form usable by plants.
- ☐ B break down organic matter and return nutrients to the ecosystem.
- ☐ C kill harmful micro-organisms.
- ☐ D can photosynthesise.

- 6** Animals which make burrows in the soil can sometimes help farmers by making air holes in the soil and mixing layers of soil. Which of these burrowing animals would a farmer like to have in her fields?

- ☐ A Mice
- ☐ B Rabbits
- ☐ C Foxes
- ☐ D Earthworms

- 7** In the food pyramid shown, each stage passes 10% of energy available to the next stage.



How many joules (J) of energy would be available to the eagle if there were 1000 MJ present in the grass?

- ☐ A 100 MJ
- ☐ B 10 MJ
- ☐ C 1 MJ
- ☐ D 100 kJ

8 All food chains:

- ☐ **A** begin with green plants.
- ☐ **B** involve humans.
- ☐ **C** start with a producer organism.
- ☐ **D** have organisms becoming more numerous, moving up the chain.

Questions 9–10 refer to the following information.

DDT (Dichloro-Diphenyl-Trichloroethane) was first used on a large scale in the 1940s to control the numbers of mosquitoes and other insects that were spreading diseases such as malaria and typhus.

By the 1960s, it was used in large amounts in agriculture because it was such an efficient poison for many arthropods. However, DDT was found to become more concentrated as it passed through the food chain and could cause illness (or death) in animals at the top of the food chain.

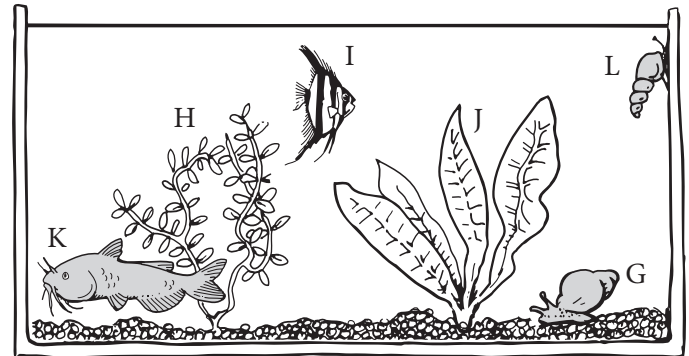
9 DDT:

- ☐ **A** is a herbicide.
- ☐ **B** is an insecticide.
- ☐ **C** kills bacteria that cause disease.
- ☐ **D** is a fertiliser.

10 What is the name of the process where the concentration of a substance builds up as it passes along a food chain?

- ☐ **A** Biodegradation
- ☐ **B** Bioaccumulation
- ☐ **C** Biomagnification
- ☐ **D** Biocompatibility

Questions 11–16 refer to the following information.
The aquarium shown (not to scale) contains six major groups of organisms (G to L) and is lit by an external light source.



11 Which of these groups contains only producers?

- ☐ **A** H, I, J, K
- ☐ **B** G, H, K
- ☐ **C** H, J
- ☐ **D** G, K

12 Which of these groups contains only consumers?

- ☐ **A** G, K, L, I
- ☐ **B** G, H, K
- ☐ **C** H, J
- ☐ **D** G, K, I

13 Which of these groups contains only organisms that photosynthesise?

- ☐ **A** G, K, L
- ☐ **B** G, H, K
- ☐ **C** H, J
- ☐ **D** K, I

14 Which type of organism (not shown in the diagram) must be present in order for materials to be recycled?

- ☐ A Consumers
- ☐ B Predators
- ☐ C Decomposers
- ☐ D Parasites

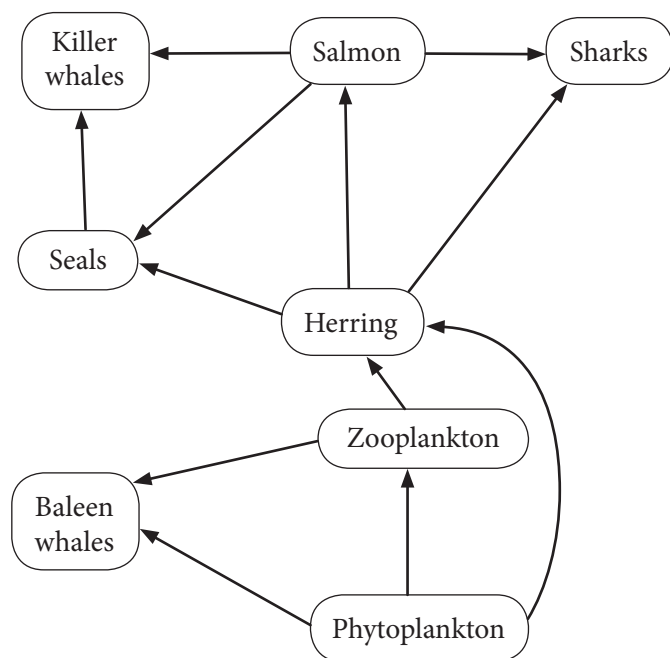
15 The aquarium and the organisms present are best described as:

- ☐ A an ecosystem.
- ☐ B a habitat.
- ☐ C a community.
- ☐ D a food chain.

16 The system shown can be described as aquatic. This means that:

- ☐ A lots of different organisms are present.
- ☐ B all organisms present breathe oxygen.
- ☐ C light is needed to provide energy.
- ☐ D water is involved.

Questions 17–20 refer to the food web below.



17 The total mass of herring would be:

- ☐ A greater than the total mass of salmon.
- ☐ B greater than the total mass of phytoplankton.
- ☐ C less than the total mass of seals.
- ☐ D unrelated to the total mass of phytoplankton.

18 Which is a primary consumer?

- ☐ A Phytoplankton
- ☐ B Baleen whales
- ☐ C Killer whales
- ☐ D Sharks

19 If the numbers of herring suddenly decreased, then the numbers of:

- ☐ A phytoplankton would decrease.
- ☐ B seals would increase.
- ☐ C sharks would decrease.
- ☐ D zooplankton would decrease.

20 All of the organic matter in the system shown will eventually be broken down by decomposer organisms. The products of this decomposition:

- ☐ A have much less mass than the original living organisms.
- ☐ B are totally lost to the system.
- ☐ C can re-enter the system by means of the phytoplankton.
- ☐ D are more complex chemicals than the original living organisms.

21 Bush flies and midges lay their eggs in the faeces of cattle. The CSIRO has imported many species of dung beetles. These insects break up the faeces of cattle. The most likely reason for the introduction of these beetles is:

- ☐ A the bush flies aren't affected by insecticides.
- ☐ B local beetles are very effective in removing cattle dung.
- ☐ C the introduction of cattle has increased the numbers of bush flies.
- ☐ D the CSIRO is the only organisation allowed to breed insects.

22 'Paterson's curse' is a plant that was introduced from the Mediterranean and has since become a serious pest. It can smother large areas of pasture, and when sprayed with herbicide it leaves bare ground which can be infested with other weeds or cause erosion. It has been suggested that biological control might be a more efficient way to limit the amount of this weed. One method of control would be to introduce a predator that would eat these plants. The major problem with introducing such a predator is that:

- ☐ A it would be very expensive.
- ☐ B it may prefer other, more useful plants and become a pest itself.
- ☐ C some people like Paterson's curse as an ornamental plant.
- ☐ D firms selling insecticides would suffer.

23 Trophic levels are the:

- ☐ A altitude at which an organism grows best.
- ☐ B feeding positions in a food chain.
- ☐ C numbers of organisms in a link of a food chain.
- ☐ D hot areas near the Equator.

24 A supermarket plastic carrier bag was described as biodegradable. This means that it:

- ☐ A is made of natural products.
- ☐ B is broken down by the wind, sun and water.
- ☐ C is broken down by bacteria.
- ☐ D will not decompose.

25 Koalas are very particular about the type of gum leaves they eat. The area around the place where a population of koalas lived was cleared. This meant that the koalas became isolated and so only had access to a limited number of trees. What would be the most likely outcome for this population of koalas competing for a limited supply of gum leaves?

- ☐ A The numbers of koalas would decrease
- ☐ B The numbers of koalas would increase.
- ☐ C The koalas would start to eat other leaves.
- ☐ D The trees would grow extra leaves.

26 In which row is the type of environment correctly matched with its description?

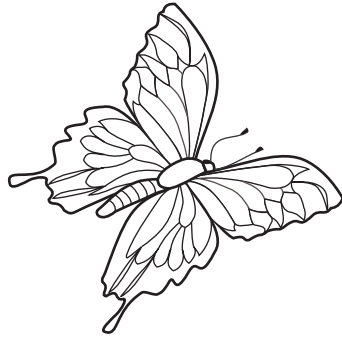
	Environment			
	Ground	Atmosphere	Water	Trees
A	terrestrial	arid	aquatic	arboreal
B	terrestrial	windy	arctic	wooden
C	earthy	humid	transparent	wooden
D	terrestrial	aerial	aquatic	arboreal

- ☐ A A ☐ B B ☐ C C ☐ D D

27 *All the members of a particular species (related organisms) found in an area.* What does this define?

- ☐ A population
- ☐ B relationships
- ☐ C biosphere
- ☐ D biome

Questions 1–8 refer to the information below.



Researchers discovered two species of insects inhabiting the same ecosystem, comprising mainly light coloured trees. The insects were very similar in all aspects apart from their colour. Species X was light coloured and species Y was dark in colour.

The researchers suspected that the colour of the insects might influence their numbers. Their idea was that the darker coloured insects would stand out against the light background of trees and hence it would be easier for predators to see and eat them.

To test this, they decided to use a 'capture, release and recapture' method. In this, the insects are captured, marked and then released. After an interval, the same area is used to trap more of the same type of insect. The researchers carried out this process and captured more light coloured insects than dark coloured insects.

1 Both light coloured and dark coloured butterflies are found in the same ecosystem. This is a:

- ☐ A conclusion.
- ☐ B fact.
- ☐ C hypothesis.
- ☐ D forecast.

2 The suspicion of the researchers that the colour of these insects influences the numbers present is a:

- ☐ A conclusion.
- ☐ B fact.
- ☐ C theory.
- ☐ D forecast.

3 The process of 'capture, release and recapture' is:

- ☐ A a theory.
- ☐ B an experiment.
- ☐ C data.
- ☐ D an observation.

4 The number of insects captured and counted is:

- ☐ A irrelevant.
- ☐ B data.
- ☐ C an interpretation.
- ☐ D proof.

5 The researchers decided that more work must be carried out in order to come to a justifiable conclusion. Which would be the most scientific thing to do next?

- ☐ A Remove the predators of the insects from the ecosystem.
- ☐ B Record the feeding habits of the predators.
- ☐ C Plant dark coloured trees.
- ☐ D Introduce a third species of insect into the ecosystem.

The researchers decided to analyse their results using the formula:

$$N = \frac{n_1 \times n_2}{M}$$

N = Estimate of total population size

n_1 = Total number of animals captured on the first visit

n_2 = Total number of animals captured on the second visit

M = Number of animals captured on the first visit that were then recaptured on the second visit

The numbers of insects are shown in the table.

	n_1	n_2	M
Light colour	51	60	20
Dark colour	30	43	15

6 What is the estimated population of the light coloured insects?

- A** 131
- B** 153
- C** 170
- D** 3 040

7 What is the estimated population of the dark coloured insects?

- A** 86
- B** 88
- C** 106
- D** 398

8 A reasonable conclusion for the experiment is that:

- A** it proved the idea that there was a relationship between colour and numbers present.
- B** it supported the idea that there was a relationship between colour and numbers present.
- C** it proved that predators ate more dark coloured than light coloured insects.
- D** light coloured insects are harder to catch than dark coloured insects.

Questions 9–10 refer to the following information.

The scientists also measured the maximum body length of the insects (to the nearest 5 mm) as shown in the table below.

	Body length (mm)				
	5	10	15	20	25
Light	2	10	23	14	2
Dark	0	8	17	5	0

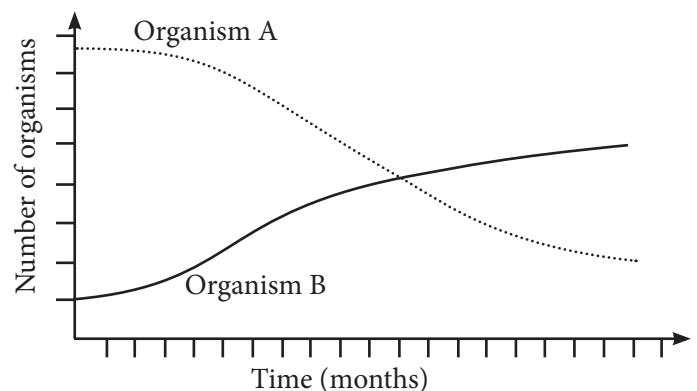
9 What was the average body length of the light insects (to the nearest mm)?

- A** 7.85 mm
- B** 10 mm
- C** 15 mm
- D** 34 mm

10 What was the average body length of the dark insects (to the nearest mm)?

- A** 7.85 mm
- B** 10 mm
- C** 14 mm
- D** 15 mm

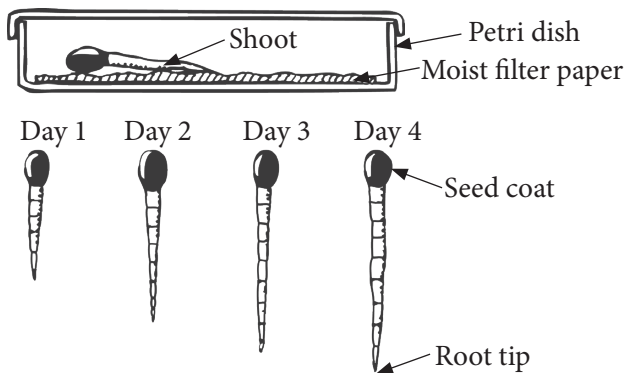
11 Organisms of type A and type B lived in the same ecosystem.



From the graph above, what is the most likely reason for the changes in numbers of organisms?

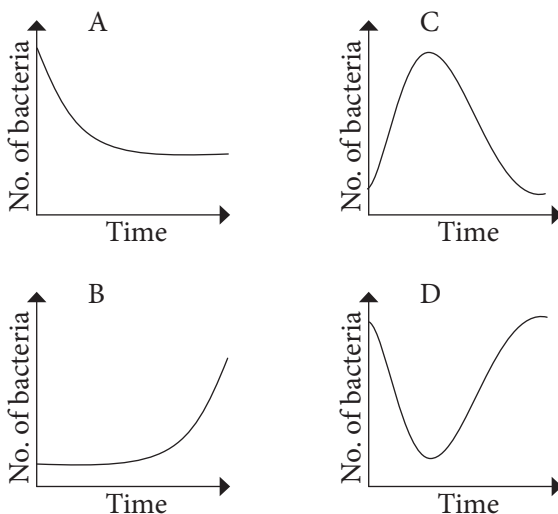
- A** The total amount of food suddenly increased.
- B** The organisms mutated.
- C** Organism A feeds on organism B.
- D** Organism B feeds on organism A.

- 1** A student took a seed that had started to put out a shoot and marked the root at equal intervals from the root tip towards the seed coat. The seedling was then placed in a Petri dish as shown and the shoot examined each day and its growth recorded.



The best conclusion that can be drawn from the data is:

- ☐ A seeds need water to germinate.
 - ☐ E shoots grow from the seed coat.
 - ☐ G shoots grow most at the root tip.
 - ☐ D shoots grow best on paper.
- 2** A researcher placed a sample of bacteria into a sealed environment with a limited amount of food, water and oxygen. He found that the number of live bacteria present initially increased rapidly, reached a maximum and then decreased.



Which graph best represents his findings?

- ☐ A A
- ☐ B B
- ☐ C C
- ☐ D D

- 3** A team of botanists was given the task of reclaiming an abandoned industrial site contaminated with waste. Their major objective was to grass the area and its surroundings so that it could eventually be used as a recreational area.

Their first discovery was that over the ten years since activities had ceased, grasses were already growing over the contaminated area and all around the site. The botanists collected grass seed from the contaminated area and also from similar but uncontaminated soil a few kilometres away.

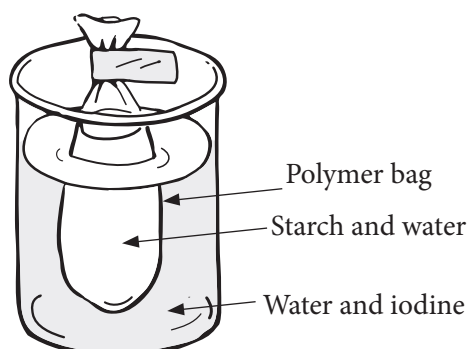
They tested the germination of both seed samples in both types of soil. The percentage of germination is shown in the table.

	Uncontaminated seed	Contaminated seed
Uncontaminated soil	97%	85%
Contaminated soil	93%	0.01%

These results illustrate that the major factor in germination of grass seeds in this situation is:

- ☐ A contamination of seeds.
 - ☐ B contamination of soil.
 - ☐ C how long seeds are left before collection.
 - ☐ D the type of soil used.
- 4** Botanists study:
- ☐ A pollution.
 - ☐ B plants.
 - ☐ C sport.
 - ☐ D industry.

- 5** A student filled a thin polymer bag with a mixture of starch and water and sealed the bag. He then placed the bag and its contents into a solution of iodine.



The bag was permeable to water and iodine but not to starch. If iodine comes into contact with starch a deep blue-black colour appears. After a short time, the contents of the bag started to turn blue-black. The most likely explanation for this is that:

- ☐ A starch is moving out of the bag.
- ☐ B water is moving into the bag.
- ☐ C iodine is moving into the bag.
- ☐ D iodine is moving out of the bag.

Questions 6–7 refer to the information below.

A farmer tested three different fertilisers on her wheat crop. She repeated this on different fields. The results are shown in the table.

	Yield of wheat (tonnes per hectare)			
Fertiliser	Field 1	Field 2	Field 3	Field 4
None	0.872	0.915	1.24	1.11
Growzem	1.34	0.883	1.81	1.54
Bettawheat	1.55	1.51	1.56	0.984
Enpeekay	1.88	0.789	1.53	1.77

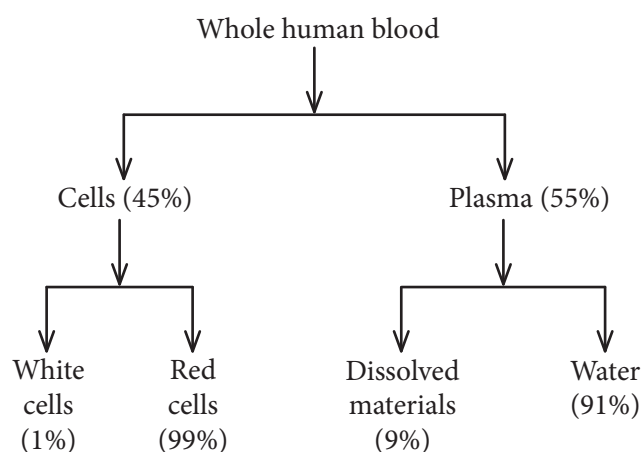
- 6** Which fertiliser had the highest average yield?

- ☐ A None
- ☐ B Growzem
- ☐ C Bettawheat
- ☐ D Enpeekay

- 7** What is the scientific name for the process where no fertiliser is applied?

- ☐ A median.
- ☐ B mean.
- ☐ C control.
- ☐ D fallow period.

- 8** Examine the diagram below, which shows the contents of whole human blood.



Water makes up what percentage of whole human blood?

- ☐ A 50%
- ☐ B 55%
- ☐ C 82%
- ☐ D 91%

- 9** Blood has an average density of 1.06 g mL^{-1} . Assuming that the average adult has 5.0 L of blood, what is the mass of blood present?

- ☐ A 53 g
- ☐ B 472 g
- ☐ C 530 g
- ☐ D 5.3 kg

ANSWERS

Topic 1 – Living things and classification

- 1 A 2 A 3 D 4 D 5 C
6 A 7 A 8 C 9 A 10 A
11 C 12 C 13 B 14 A 15 D
16 C 17 D 18 B 19 C 20 A
21 B 22 D 23 D 24 C 25 C
26 C 27 B 28 B 29 D

Topic 2 – Animal cells

- 1 D 2 C 3 A 4 B 5 A
6 B 7 B 8 A 9 A 10 A
11 D 12 B 13 C 14 A 15 D
16 B 17 C 18 D

Topic 3 – Plant cells

- 1 B 2 A 3 C 4 D 5 C
6 C 7 C 8 A

Topic 4 – Plants

- 1 A 2 C 3 B 4 A 5 C
6 D 7 C 8 B 9 D 10 C
11 B 12 D 13 B 14 D 15 C
16 D 17 B 18 C 19 C 20 C

Topic 5 – Diseases

- 1 A 2 B 3 B 4 D 5 B
6 C 7 D 8 D 9 C 10 D
11 D 12 A 13 A 14 B 15 C
16 A 17 A

Topic 6 – The skeletal and muscular systems

- 1 A 2 C 3 A 4 C 5 C
6 A 7 D 8 A 9 C 10 A
11 B 12 C 13 C 14 A 15 B
16 D 17 A

Topic 7 – The endocrine system

- 1 B 2 D 3 C 4 D 5 B
6 C 7 A 8 B 9 A 10 B
11 C 12 D 13 C 14 C 15 B
16 D 17 B 18 C 19 B

Topic 8 – The nervous system

- 1 B 2 A 3 B 4 C 5 D
6 A 7 B 8 D 9 A 10 C
11 A 12 B 13 B 14 A 15 D
16 C 17 A 18 C

Topic 9 – The circulatory system

- 1 B 2 B 3 D 4 B 5 D
6 B 7 B 8 A 9 C 10 D
11 C 12 B 13 D 14 A 15 A
16 C 17 B 18 D 19 D 20 A
21 A

Topic 10 – The respiratory system

- 1 D 2 A 3 C 4 B 5 D
6 C 7 D 8 A 9 B 10 A
11 C 12 B 13 D 14 A 15 B

Topic 11 – The senses

- 1 B 2 D 3 C 4 A 5 B
6 B 7 A 8 D 9 A 10 A
11 B 12 C 13 D 14 C 15 D
16 B 17 A 18 A 19 D 20 B
21 C 22 B 23 B 24 D 25 C
26 D 27 B 28 D 29 A 30 D
31 D 32 B

Topic 12 – Human reproduction

- 1 C 2 A 3 C 4 A 5 A
6 B 7 B 8 C 9 A 10 D
11 C 12 C 13 A 14 C 15 B
16 A 17 C 18 C 19 D 20 C
21 B 22 D 23 A 24 C 25 B
26 A 27 C 28 C 29 D 30 C
31 A 32 B 33 A 34 A 35 A
36 A

Topic 13 – Genetics

- 1 B 2 D 3 C 4 B 5 D
6 A 7 D 8 C 9 B 10 B
11 A 12 B 13 C 14 D 15 C
16 B 17 D 18 B 19 B 20 A

Topic 14 – Balance in the blood

- 1 D 2 C 3 B 4 C 5 A
6 D 7 A 8 C 9 A 10 A

Topic 15 – Evolution

- 1 D 2 D 3 C 4 C 5 A
6 A 7 B 8 D 9 C 10 A
11 B 12 C 13 D 14 B 15 A
16 B

Topic 16 – Ecology

- 1 B 2 A 3 A 4 C 5 B
6 D 7 C 8 C 9 B 10 C
11 C 12 A 13 C 14 C 15 A
16 D 17 A 18 B 19 C 20 C
21 C 22 B 23 B 24 C 25 A
26 D 27 A

Topic 17 – Populations

- 1 B 2 C 3 B 4 B 5 C
6 B 7 A 8 B 9 C 10 D
11 D

Topic 18 – Interpretation of data

- 1 C 2 C 3 A 4 B 5 C
6 D 7 C 8 A 9 D

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The senses
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Interpretation of data

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